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"CGLOPS-1"
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VALIDATION REPORT

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List of Acronyms

A

AD.....Applicable Document, Applicable Document
ASCAT.....Advanced Scatterometer
ATBD.....Algorithm Theoretical Baseline Document

C

CEOS.....Committee on Earth Observation Satellites
CGLOPS.....Copernicus Global Land Operations
CLMS.....Copernicus Land Monitoring Service
COG.....Cloud Optimized GeoTIFF
cov.....Covariance

E

ECMWF.European Centre for Medium-Range Weather Forecasts
EUMETSAT.....European Organisation for the Exploitation of Meteorological Satellites

G

GCOS.....Global Climate Observing System
GLDAS.....Global Land Data Assimilation System

I

INSPIRE.....Infrastructure for Spatial Information in Europe
ISMN.....International Soil Moisture Network

L

LPV.....Land Product Validation
LSM.....Land Surface Model
LST.....Land Surface Temperature

N

NASA.National Aeronautics and Space Administration
NETCDF.....Network Common Data Format
NOAH.....acronym referring to the organizations that developed it: NCEP (National Centers for Environmental Prediction), Oregon State University, Air Force, and Hydrologic Research Lab.
NRT.....Near Real Time

P

PQAD.....Product Quality Assurance Document
PUM.....Product User Manual, Product User Manual

Q

QAR.....Quality Assessment Report
QFLAG.....Quality Flag
QM.....Quality Monitoring

R

rho.....Pearson Correlation

S

SQUAD.....System Quality Assurance Document
SSF.....Surface State Flag
SSM.....Surface Soil Moisture
std.....Standard deviation
SWI.....Soil Water Index

U

ubRMSD.....unbiased Root Mean Square Difference

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EXECUTIVE SUMMARY

The Copernicus Land Monitoring Service (CLMS) produces a series of qualified bio-geophysical products on the status and evolution of the land surface. The products are used to monitor vegetation, crops, water cycle, energy budget and terrestrial cryosphere. Production and delivery of the parameters take place in a timely manner and are complemented by the constitution of long-term time series.

This document describes the scientific Quality Monitoring (QM) for the Soil Water Index (SWI) in accordance to the System Quality Assurance Document [SQAD].

Due to the introduction of a new major product version (from v3 to v4.0.1) in 2025, the document contains both the annual QM for SWI v4.0.1 for the year 2025 as well as a validation of the reprocessed long-term archive (2007-2025) for v4.0.1 and a comparison of the different data versions for the overlap-period (2007-2024).

The aim of the annual global analysis is to assess the performance of the near real time (NRT) SWI v4.0.1 product in the period from 2025-01-01 to 2025-12-31, relative to the performance of the SWI archive from 2007-2024. The validation presented in this report is performed according to the procedures defined in the Product Quality Assurance Document [CGLOPS1_PQAD].

For the global analysis, SWI time series are compared to soil moisture data provided by the Global Land Data Assimilation System GLDAS-NOAH V2.1 (Beaudoing and Rodell, 2020). The SWI for T-values 1 and 20 days (T: characteristic time length) are hereby compared to soil-moisture simulations at a depth of 0-0.1 m whereas T=100 is compared to a depth of 0.1-0.4 m.

For the in-situ comparison, SWI time series are compared to soil-moisture sensor data provided by the International Soil Moisture Network (ISMN). Here, the values of SWI T=1 and 20 are compared to stations at a depth < 0.1m whereas values of SWI T=100 are compared to stations located at a depth of 0.5m.

For the inter-comparison of the data versions v3 and v4.0.1, the SWI time series are compared for the overlap-period 2007-2024 at equal T values.

The annual validation of the SWIv4 product for 2025 shows a stable good performance against both model data and in-situ data. Expected weak performances are observed in subarctic regions, densely vegetated areas (tropics), deserts, and areas that show high topographic complexity.

In quantitative terms on a global view, generally more than 90% of the grid points fulfil the threshold requirement for SWIv4 when compared to GLDAS NOAH whereas the goal is fulfilled by 42-61% (period 2007-2024) or 31-37% (for 2025), respectively. Lower agreement

was observed in the comparison to in-situ data: while figures are still above 90% with respect to the threshold requirement, the goal is reached by 16-34% (period 2007-2024) and 28-45% for 2025.

The comparison of SWIv3 against SWIv4 indicates a similar performance with overall improved metrics found for SWIv4 against both model and in-situ data.

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1 BACKGROUND OF THE DOCUMENT

1.1 SCOPE AND OBJECTIVES

This document presents the results of the quality assessment and validation of the operational SWI v4.0.1 product as well as a comparison of the SWI v3 dataset against SWI v4.0.1.

The comparison of SWI v3 with SWI v4.0.1 is performed with respect to the overlap period from 2007-01-01 to 2024-12-31. The primary objective of the version comparison is to inform users about expected differences in the data characteristics associated with the major version change and to assess effects of the changed input data stream on obtained performance metrics.

The annual quality evaluation is performed on SWI data for the year from 2025-01-01 to 2025-12-31 with a reference period from 2007-01-01 to 2024-12-31. The objective of the annual quality evaluation is to check that the operational products keep the same level of quality in the period under study compared to the reference period.

1.2 CONTENT OF THE DOCUMENT

This document is structured as follows:

- Chapter 2 recalls the requirements.
- Chapter 3 describes the methodology for quality assessment and the reference datasets.
- Chapter 4 presents the results of the analysis.
- Chapter 5 discuss the main outcome and summarizes the conclusions of the study.
- Chapter 6 gives some recommendations and perspectives.

1.3 RELATED DOCUMENTS

1.3.1 Applicable documents

AD1: Part 2: Technical specifications of Framework Service Contract – Operation of the bio-geophysical variables systematic monitoring of the Global Land Component of the Copernicus Land Service 'CGLOPS' JRC/2023/OP/0273, 19th April 2023.

Available at https://etendering.ted.europa.eu/cft/cft-display.html?cftId=13795

1.3.2 Input

Document ID	Descriptor
CGLOPS1_PQAD	Product Quality Assurance Document of the Global Operations of the Copernicus Land Monitoring Service.
CGLOPS1_SQAD	System Quality Assurance Document of the Global Land Service.
CGLOPS1_ATBD_ SWI0.1	Algorithm Theoretical Baseline Document – Soil Water Index 0.1°

Available at: https://land.copernicus.eu/en/technical-library

1.3.3 Output

Document ID	Descriptor
CGLOPS1_PUM_SWI0.1	Product User Manual – Soil Water Index 0.1°

Available at: https://land.copernicus.eu/en/technical-library

1.3.4 External documents

Document ID	Descriptor
GCOS#245	The 2022 GCOS ECVs Requirements

Available at: https://library.wmo.int/records/item/58111-the-2022-gcos-ecvs-requirements-gcos-245?offset=1

2 REQUIREMENTS

Following the applicable document [AD1], the requirements relevant for SWI are described below.

2.1 SPECIFIC TECHNICAL DETAILS AND REQUIREMENTS

SWI – Soil Water Index	
The surface soil water index provides daily information about moisture conditions in different soil depths. Soil moisture is a very heterogeneous variable and varies on small scales with soil properties and drainage patterns. Satellite measurements integrate over relative large-scale areas, with the presence of vegetation adding complexity to the interpretation. The SWI quantifies the amount of water (m^3/m^3) in soil layers at various depths and is derived from the SSM using a water balance model. The freeze and thaw status of soil is provided.	
Geometric Properties:	
spatial resolutions	3-10km/1km
geolocation precision	Better than 0.5 pixels
coordinate position	Centre of the pixel
geodetical datum	WGS84
geographic projection	regular latitude/longitude grid
Geographic Coverage:	Global/EU
Time span of output product:	SWI: Daily SWI10: 10-day period SWI TS (time series): 6 months
Timeliness of output delivery:	SWI: 12hrs SWI10: within 3 days after synthesis period
Uncertainty	Threshold: 0.1 m^3/m^3 Goal: 0.04 m^3/m^3
Stability	Threshold: 5% Goal: 3%

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2.2 FURTHER REQUIREMENTS

2.2.1 Output product composition

Products may contain various information layers and ancillary information – the base reference for product packages are the operational products as on 01.03.2023.

2.2.2 Data Structure

Data coding[1] shall be compatible with the Global Land products as on 01.03.2023 and/or follow the INSPIRE specifications, where applicable.
Ancillary information shall be as currently used and include at least the following:

- The number of measurements per pixel used to generate any synthesis product
- The per-pixel date of the individual measurement or the start-end dates of the period covered
- Quality indicators, with explicit per-pixel identification of the cause of an anomalous parameter result.

The product naming and filename conventions that are used in the Copernicus Global Land component production as on 01.03.2023 shall be followed. This may be adapted for complete product collections upon agreement with the contracting authority during the Framework.

2.2.3 Data format

To ensure interoperability with the current Global Land component (operational product data formats and archive data formats) and other Copernicus services, all datasets will be available in NETCDF. Additional format such as Cloud Optimized Geotiff (COG) or ZARR format can be proposed for production or could be requested by the contracting authority during the Framework contract.

2.2.4 Uncertainties and Validation

Uncertainties indicated in the product specifications above follow the threshold and goal proposed by GCOS#245.

Uncertainties estimates should account for the error propagation uncertainty coming from input data though the retrieval algorithms as during the contract period, ESA plans to imbed uncertainties in the Sentinel-3 ground segment products and, for Sentinel-2, there is an offline

[1] Data coding is the provision for the number of bits, range of values, usage of reserved values, content of status map, conversion to physical values formulas.

tool to determine Level 1 uncertainties; these can be used for propagation in the production chain.

Validation of the products shall conform to at least the CEOS LPV standards. Wherever appropriate the bio-geophysical variables shall be validated and compared to CEOS CAL/VAL data sets and/or Ground-Based Observations for Validation (GBOV) of Copernicus Global Land component when biophysical parameters are available.

2.2.5 Input data

Copernicus sentinel data are available from https://dataspace.copernicus.eu

Bio-geophysical variable products should be based on common base reflectance data:

- Sentinel 3: Derived Top-of-Canopy reflectance may be brokered or produced as it is the case on 01.03.2023 under the CGLOPS contracts;
- Sentinel 2: the Global Land Sentinel-2 Global Mosaic (S2GM)[2] component provides temporal mosaic of surface spectral bands that can be brokered and/or directly used.

Up to and including 2019, products archives of the Global Land component have been based on SPOT VGT, Proba-V, ENVISAT, MODIS, TOPEX/Poseidon, Jason-1, Jason-2, Jason-3, datasets, which are available through the https://dataspace.copernicus.eu.

Land Surface Temperature (LST) and SWI can be based on geostationary and other satellite data.

Ancillary satellite data that is purchased through Copernicus and put at disposal of the Services is available through the Data Warehouse and will become available on the Copernicus Dataspace Ecosystem.

Ancillary data sets, other than satellite imagery described above, that might be required shall be the responsibility of the contractor.

2.2.6 Product delivery

Products shall be delivered to the Copernicus Land Component dissemination.
The Copernicus Data Space Ecosystem infrastructure will be used to provide access to the final map products.

[2] https://land.copernicus.eu/en/products/global-image-mosaic

3 QUALITY ASSESSMENT AND VALIDATION METHOD

3.1 SWI PRODUCTS UNDER ANALYSIS

The SWI0.1° products ingest 12.5 km Surface Soil Moisture (SSM) observations from Metop ASCAT, as derived by EUMETSAT's HSAF. The SWI output in CGLOPS has global coverage. The version 3, derived from the H104 HSAF product, was provided from January 2007 up to July 13th, 2025. HSAF terminated production of H104 at this point. Consequently, from 14th July 2025, the SWI 0.1° version 4 was derived from the HSAF product H29. The new ASCAT data streams include model enhancements like long-term trend correction and adaptive vegetation correction. While improving the quality of the ASCAT SSM product, it also had the consequence that the CLMS archive (v3.2.1) and the CLMS NRT products (v4.0.1) were no longer consistent, and a reprocessing activity had to be initiated.

Details on the HSAF product updates can be found at:

- https://user.eumetsat.int/news-events/news/changes-to-ascat-soil-moisture-products
- https://hsaf.meteoam.it/Products/Detail?prod=H29

Irrespective of the version, the daily SWI 0.1 processing calculates the Soil Water Index for eight different characteristic time lengths T (namely: 1, 5, 10, 15, 20, 40, 60 and 100 days) together with the respective quality flags (QFLAG) for all 3-minute input files of the target day. The QFLAG describes the level of confidence that the SWI calculation has had sufficient data, of acceptable quality, within the interval T to derive a meaningful SWI value.

Soil moisture retrieval from scatterometer data has certain limitations. Soil moisture cannot be retrieved when the surface is frozen or covered by snow, dense vegetation, or water. Snow and frozen surface conditions are dynamic factors and should be flagged carefully to avoid miscalculation of SWI. In SWI product versions before 4.0, a numerical algorithm developed at TU Wien was used for detection of freeze/thaw conditions based on the parameters and thresholds extracted by processing of the ASCAT normalized backscatter time series and using distinctive decision trees. A Surface State Flag (SSF) was attached to each backscatter measurement which indicated frozen/unfrozen surface, thaw condition/snow melt. In product version 4.0, meteorological forecasts from ECMWF were incorporated, now providing the basis for the SSF flag. ECMWF provides forecasts for skin temperature and snow depth every 2 hours. Similar to the advisory flag of the H29 Metop ASCAT Surface Soil Moisture Near Real Time 12.5 km product, binary frozen soil and snow cover masks are derived from forecasts of skin temperature and snow depth. Since the SSF is no longer a satellite-derived quantity, a dedicated validation of the SSF itself is not required here.

3.2 OVERALL PROCEDURE

For the SWI validation, the following performance metrics are evaluated:

- **Pearson Correlation coefficient (rho):**

$$rho = \frac{cov(X,Y)}{std(X)\ std(Y)} \qquad with$$

cov() ……. Covariance
std() …….. Standard deviation

  - **p-value**
    A measure on the statistical significance of the obtained Pearson correlation evaluated by assuming that the probability density function of the correlation coefficient is represented by a beta-distribution on the interval [-1, 1] as implemented in the python library scipy (Virtanen et al. 2020).

- **(Unbiased) Root Mean Square Difference (ubRMSD):**
  The values of this statistic are tested against the "uncertainty" requirements listed in section 2.1. The term "unbiased" hereby refers to the fact that time series undergo "unit matching" prior to the evaluation of the RMSD (see below).

$$ubRMSD = \sqrt{\frac{\sum_{i=1}^{N}(x_i - y_i)^2}{N}}$$

  Before the metrics can be calculated, the following pre-processing steps are required:

1. **Spatial matching:**
   The closest datapoint in the reference dataset is identified for each grid point in the SWI dataset.

2. **T value to reference data depth matching:**
   Depending on the used SWI T-value, an appropriate depth-layer in the reference dataset is selected for model and in-situ data as presented in Table 1. General recommendations and insights on the attribution of T-values to soil depth are given in the Product User Manual [CGLOPS1_PUM_SWI0.1]

3. **Application of SWI and reference-data flags**

- In-situ soil-moisture time series are masked with respect to the quality flag provided by the ISMN network. Only values marked as "G" (good) are considered in the validation.
- SWI time series are masked with respect to the Surface State Flag (SSF) and the Quality Flag (QFLAG) according to: $SSF \neq 2$ (i.e. not "Frozen") and $QFLAG_T > 5\%$

4. **Temporal matching:**
   The reference time series are temporally matched to the SWI observation timestamp by selecting the nearest-neighbour in time with a maximum time difference of 3 hours.

5. **Unit matching:**
   The measurement unit of the SWI data (i.e. % relative saturation) must be matched to the measurement unit of the reference datasets (i.e. m^3/m^3) in order to allow calculation of the expected goal/threshold indicators for the performance metric ubRMSD in m^3/m^3. The scaling is performed via a "mean standard deviation scaling" to transform the SWI values into the unit of the reference dataset. Note that this scaling method has no impact on the obtained Pearson correlation values.

After the metrics are evaluated, statistically insignificant results are masked according to:

$$p\text{-}value \leq 0.05$$

Even though the p-value is an estimate on the significance of Pearson correlation, the mask is also applied to ubRMSD as we do not expect ubRMSD to be meaningful if the associated correlation is insignificant.

The validation has been performed for all T-values. However, in order to keep the report concise, results are only presented for selected T-values. No distinct difference in the behaviour has been found for the other T-values.

Table 1: Matching between SWI T-values and reference-data depth layers.

SWI T-Value	GLDAS Layer depth	In-situ observation depth
1, 5, 10, 15, 20, 40	0 – 0.1 m	0 – 0.1 m
60, 100	0.4 – 1 m	0.5 m

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3.3 MODELLED REFERENCE PRODUCTS

Modelled reference products against which the SWI product is compared are taken from the GLDAS (Global Land Data Assimilation System) Noah-LSM (Land Surface Model), version 2.1. The GLDAS Noah Land Surface Model (Rodell et al., 2004, Beaudoing and Rodell, 2020) is produced by NASA. It includes four soil layers with the bottom depth at 0.1, 0.4, 1.0, and 2.0 m respectively, and simulates over 30 land surface parameters in total. Data is available with a temporal sampling of three hours on a global 0.25° grid. The model soil moisture data was converted from kg/m² to m^3/m^3 by using the formula SM [m^3/m^3] = SM [kg/m^2] * 0.001 * 1/d, where d is the thickness of the soil layer in meters. The factor 0.001 is due to the assumption that 1 kg of water represents 1000 cm^3, which is 0.001m^3.

The GLDAS model dataset is chosen for the validation because it provides consistent long-term global data, and is updated regularly at relatively small latency. The consistency of the dataset also makes it likely that future QAR exercises can use them as well. This would have been less likely if we had chosen e.g. soil moisture products derived from other remote sensing platforms that are exposed to risks lying in single systems, e.g. which are vulnerable to satellite failure.

The soil moisture component of the model is, of course, not free of errors, but there are comprehensive studies on the global quality of the modelled soil moisture. Dorigo et al. (2010) investigated the quality of the GLDAS and other models with the triple collocation method using remotely sensed soil moisture datasets as the reference datasets and found consistent performances among global modelled soil moisture datasets. Chen et al. (2013) compared the GLDAS model output against in-situ stations on the Tibetan plateau and found that the model underestimates surface soil moisture (0-5 cm) but was in good agreement with a deeper layer (10 – 40 cm). Correlations of 0.69 (0 – 5 cm) and 0.76 (10 – 40 cm) were found. Because of these results suggesting a high agreement of the models, and following review board recommendations, the GLDAS data is used for the SWI evaluation, as it also features a relatively high resolution (0.25°) for a global model dataset, which is acceptable in respect to the SWI's resolution of 0.1°.

For the QAR 2025, the GLDAS-Noah Version 2.1 was used to cover the period from 2007-01-01 to 2025-11-30 and due to the delayed delivery of the GLDAS dataset, the GLDAS-Noah Early Product Version 2.1 was used as extension for the last month of the validation period from 2025-12-01 to 2025-12-31.

3.4 IN-SITU REFERENCE PRODUCTS

For the in-situ comparison, a subset of soil-moisture stations provided by the International Soil Moisture Network (ISMN; Dorigo et al., 2011, 2013, 2021) is used, selected by the following criteria:

- The station must be located at a depth according to Table 1
- The overlap-period between SWI and the in-situ soil-moisture time series must cover at least
    - 100 days of overlapping measurements for single-year comparison
    - 300 days of overlapping measurements for full timespan comparison

Table 2 summarizes all ISMN stations that met the above criteria for the validation period. The "First / Last Year" column hereby reports the year of the first/last timestamp in the in-situ time series and "Days" reports the effective number of days with valid comparison data for the comparison of SWI T=1 against ISMN stations with depth <0.1m. Note that the figures in Table 2 do not refer to the history of the network as such but represent ranges observed for the individual stations in the network with different installation dates, dismantlement dates,…

The geographical locations of networks are displayed in Figure 28. For detailed information on each network, please refer to the references provided in Section 7.1.

Table 2: ISMN stations used for in-situ validation.

ISMN Network	# Stations	# Days	First Year	Last Year	URL
SNOTEL	1363	100 - 2958	1999 - 2024	2007 - 2026	http://www.wcc.nrcs.usda.gov/
SCAN	590	100 - 4234	1997 - 2024	2009 - 2026	http://www.wcc.nrcs.usda.gov/
USCRN	275	101 - 3522	2009 - 2012	2016 - 2026	https://www.ncei.noaa.gov/access/crn/
COSMOS-UK	191	102 - 361	2013 - 2019	2016 - 2023	https://cosmos.ceh.ac.uk/
SOILSCAPE	188	101 - 720	2011 - 2024	2012 - 2026	http://soilscape.usc.edu/
ARM	162	104 - 590	1996 - 2020	2007 - 2026	http://www.arm.gov/
PBO-H2O	158	100 - 1807	2006 - 2015	2013 - 2017	https://gnss-reflections.org/maps?product=smc
RISMA	148	107 - 1118	2013 - 2015	2017 - 2020	https://agriculture.canada.ca/en/agricultural-production/soil-and-land/real-time-situ-soil-monitoring-agriculture
GROW	144	100 - 395	2017 - 2018	2018 - 2019	https://growobservatory.org/index.html
COSMOS	137	103 - 2223	2009 - 2015	2010 - 2020	http://cosmos.hwr.arizona.edu/
TWENTE	130	108 - 1607	2008 - 2019	2009 - 2020	https://doi.org/10.5194/essd-15-1889-2023
CTP-SMTMN	106	301 - 512	2010 - 2012	2015 - 2016	
HOAL	96	107 - 1107	2013 - 2019	2019 - 2021	https://hoal.hydrology.at/research/soil-moisture
OZNET	87	110 - 3299	2001 - 2006	2010 - 2021	http://www.oznet.org.au/

HOBE	73	100 - 293	2009 - 2013	2011 - 2019	http://www.hobe.dk/
SMN-SDR	70	130 - 183	2018	2019	https://doi.org/10.11888/Soil.tpdc.271425
SMOSMANIA	68	121 - 355	2007 - 2008	2015 - 2025	http://www.hymex.org
FMI	62	104 - 1620	2007 - 2014	2012 - 2026	http://fmiarc.fmi.fi/
MAQU	61	100 - 1002	2008 - 2018	2012 - 2019	https://www.itc.nl/about-itc/organization/scientific-departments/water-resources/earth-observation-sites/tibetan-plateau-peoples-republic-of-china/
HiWATER-EH-WSN	56	102 - 116	2012	2012	http://www.westgis.ac.cn/
SONTE-China	52	145 - 386	2020	2022	
TxSON	41	155 - 1972	2014 - 2015	2019 - 2022	http://www.beg.utexas.edu/research/programs/txson
SWEX-POLAND	40	110 - 889	2000 - 2010	2009 - 2013	
UDC-SMOS	40	100 - 767	2007 - 2010	2008 - 2011	http://www.geographie.uni-muenchen.de/department/fiona/forschung/projekte/index.php?projekt_id=103
CW3E	40	101 - 832	2017 - 2024	2022 - 2026	https://cw3e.ucsd.edu/
NGARI	35	100 - 752	2010 - 2013	2014 - 2019	https://www.itc.nl/about-itc/organization/scientific-departments/water-resources/earth-observation-sites/tibetan-plateau-peoples-republic-of-china/
KIHS-CMC	31	152 - 2599	2008	2019	http://kihs.re.kr
AMMA-CATCH	28	108 - 2901	2006	2017 - 2018	http://www.amma-catch.org
SASMAS	28	163 - 330	2005 - 2006	2007	http://www.eng.newcastle.edu.au/sasmas/SASMAS/sasdata.html
BIEBRZA-S-1	28	141 - 192	2015	2018	http://www.igik.edu.pl/en
KIHS-SMC	26	142 - 216	2007	2019	http://kihs.re.kr
ROMPS	26	101 - 1662	1989 - 2014	2023 - 2026	http://sdss.caiag.kg/sdss/index.php?&page=measure_page
VDS	26	110 - 641	2017 - 2020	2018 - 2021	https://www.vandersat.com/
RSMN	25	102 - 315	2014 - 2015	2015 - 2026	http://assimo.meteoromania.ro
REMEDHUS	25	152 - 278	2005 - 2007	2010 - 2025	http://campus.usal.es/~hidrus/
BNZ-LTER	23	103 - 423	2000 - 2004	2009 - 2013	http://www.lter.uaf.edu/
ORACLE	23	147 - 1528	1996 - 2008	2013	http://gisoracle.irstea.fr/?lang=en;https://bdoh.irstea.fr/ORACLE/
XMS-CAT	22	107 - 291	2016 - 2023	2020 - 2026	https://visors.icgc.cat/mesurasols/#9/42.1765/1.1132
PTSMN	21	130 - 501	2016	2018	
Berlin	21	214 - 918	2020 - 2022	2026	
TAHMO	20	136 - 365	2019	2021 - 2022	https://tahmo.org/
UMBRIA	19	204 - 2210	2002 - 2011	2008 - 2017	http://www.cfumbria.it/; http://hydrology.irpi.cnr.it/
MySMNet	16	171 - 479	2014	2015	
TERENO	15	166 - 294	2009 - 2012	2019 - 2026	https://www.tereno.net/joomla/index.php/overview
GTK	13	109 - 425	2004 - 2006	2010 - 2012	https://www.gtk.fi/en/
FR-Aqui	13	231 - 352	2012 - 2015	2012 - 2025	
Ru-CFR	13	106 - 1002	2015	2025	
SKKU	12	112 - 230	2014 - 2015	2014 - 2017	

USDA-ARS	12	124 - 247	2002	2009	https://www.ars.usda.gov/
NAQU	11	337 - 688	2010 - 2012	2018 - 2019	https://www.itc.nl/about-itc/organization/scientific-departments/water-resources/earth-observation-sites/tibetan-plateau-peoples-republic-of-china/
FLUXNET-AMERIFLUX	11	131 - 335	2000 - 2017	2013 - 2020	http://ameriflux.lbl.gov/
MOL-RAO	10	199 - 281	2003	2016 - 2020	http://www.dwd.de/mol
IMA-CAN1	9	149 - 202	2011	2015	http://sustag.imamoter.cnr.it/index.php/cannona-db
WSMN	7	101 - 332	2011 - 2013	2016	http://www.aber.ac.uk/wsmn
DWD	7	209 - 319	2024	2026	
STEMS	6	144 - 227	2015	2023 - 2024	https://sustag.to.cnr.it/index.php/cannona-db
WATERLINE	6	264 - 310	2021	2025	https://www.chistera.eu/projects/waterline
SD-DEM	6	138 - 159	2005 - 2014	2020	http://dx.doi.org/10.7167/2013/297973
ICN	5	181 - 218	1998	2010	http://www.isws.illinois.edu/warm/
NVE	5	114 - 258	2012 - 2017	2018 - 2022	https://www.nve.no/hydrology/?ref=mainmenu
iRON	4	106 - 139	2012 - 2016	2025	https://agci.org/iron
DAHRA	3	117 - 122	2002 - 2004	2016	https://ign.ku.dk/english/research/geography/environment-society-developing-countries/
VAS	3	117 - 122	2010	2012	http://nimbus.uv.es/
LABFLUX	3	112 - 604	2018 - 2020	2024 - 2025	https://www.dist.polito.it/il_dipartimento/laboratori/labflux
IPE	2	139 - 173	2008 - 2012	2012 - 2020	
UMSUOL	2	159 - 271	2009	2017	http://www.arpa.emr.it/sim/
IIT-KANPUR	1	178	2011	2012	http://www.iitk.ac.in/
HSC-SEOLMACHEON	1	230	2008	2009	http://www.hsc.re.kr

4 RESULTS

4.1 COMPARISON BETWEEN SWIV3 AND SWIV4

This section encompasses a direct comparison of the new major SWI dataset version v4.0.1 against the precursor SWIv3 as well as a comparison of obtained performance metrics of SWIv3 and SWIv4.0.1 against GLDAS NOAH v2.1 and in-situ stations of the ISMN network.

4.1.1 Comparison of SWIv3 and SWIv4 performance metrics for GLDAS NOAH

4.1.1.1 Pearson Correlation

Figure 1 and Figure 2 show the obtained Pearson correlation values for SWI v3 and SWIv4.0.1 T=1 against GLDAS NOAH soil moisture 0-0.1m and Figure 3 shows the associated difference.

An overall similar performance is obtained for SWIv4.0.1 compared to SWIv3 with slightly improved correlations in Europe, the United States, Canada and Russia and slightly lower correlations in South America, the Middle East, and South-East Asia.

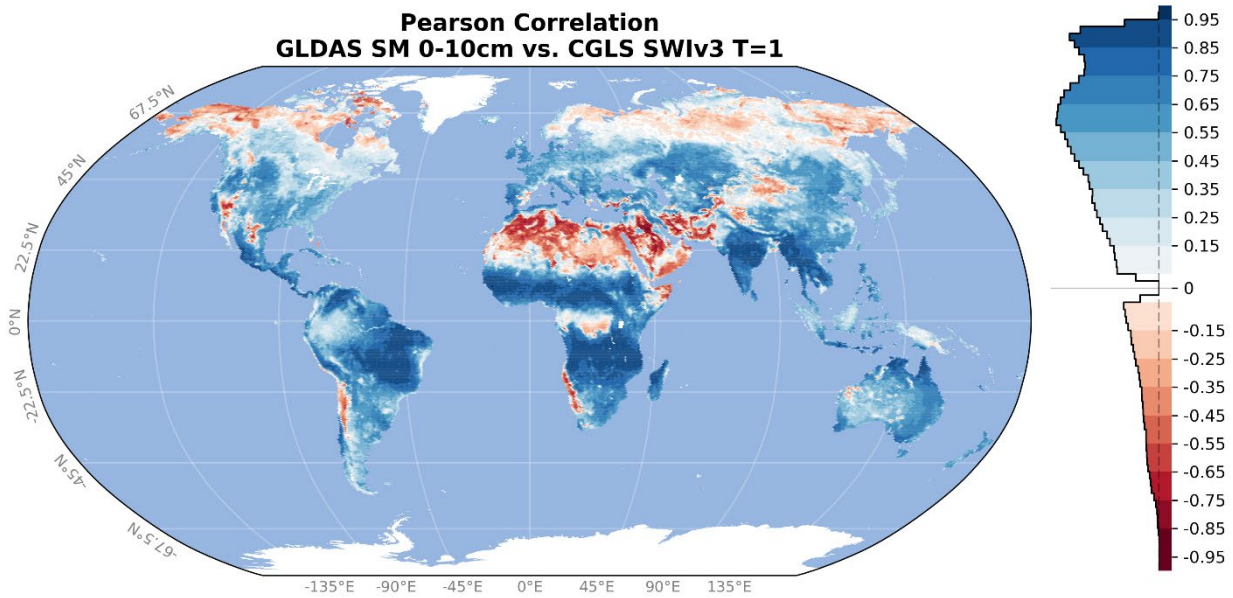


Figure 1: Pearson correlation of SWIv3 T=1 vs. GLDAS NOAH soil moisture 0-0.1m for the comparison validation period (2007-2024)

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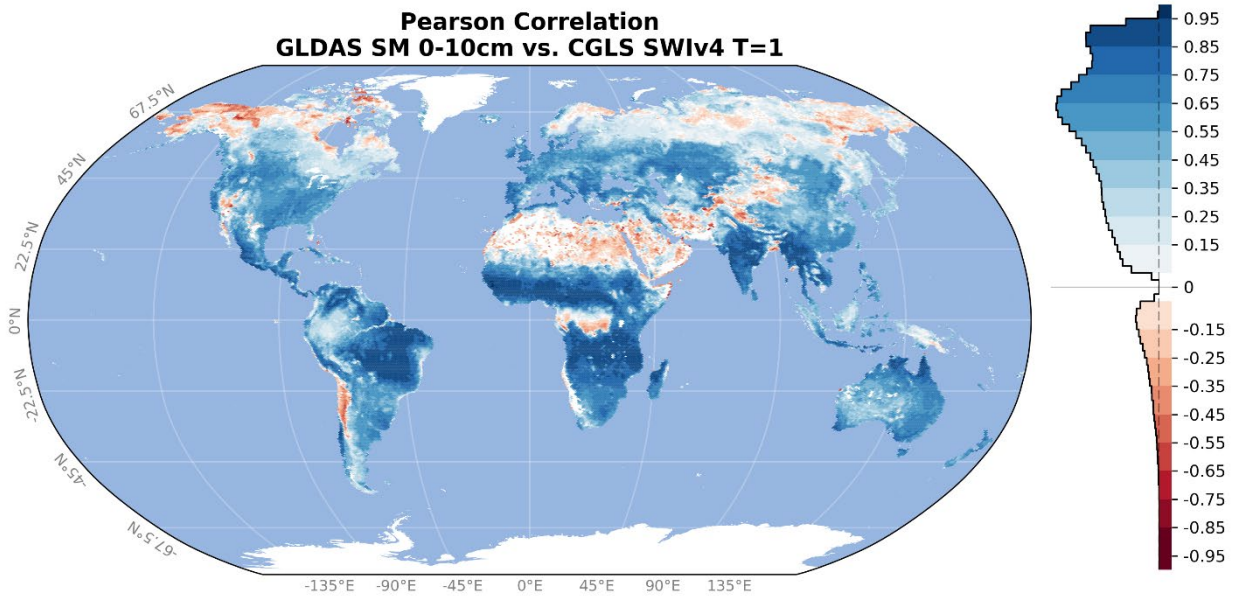


Figure 2: Pearson correlation of SWIv4 T=1 vs. GLDAS NOAH soil moisture 0-0.1m for the comparison validation period (2007-2024)

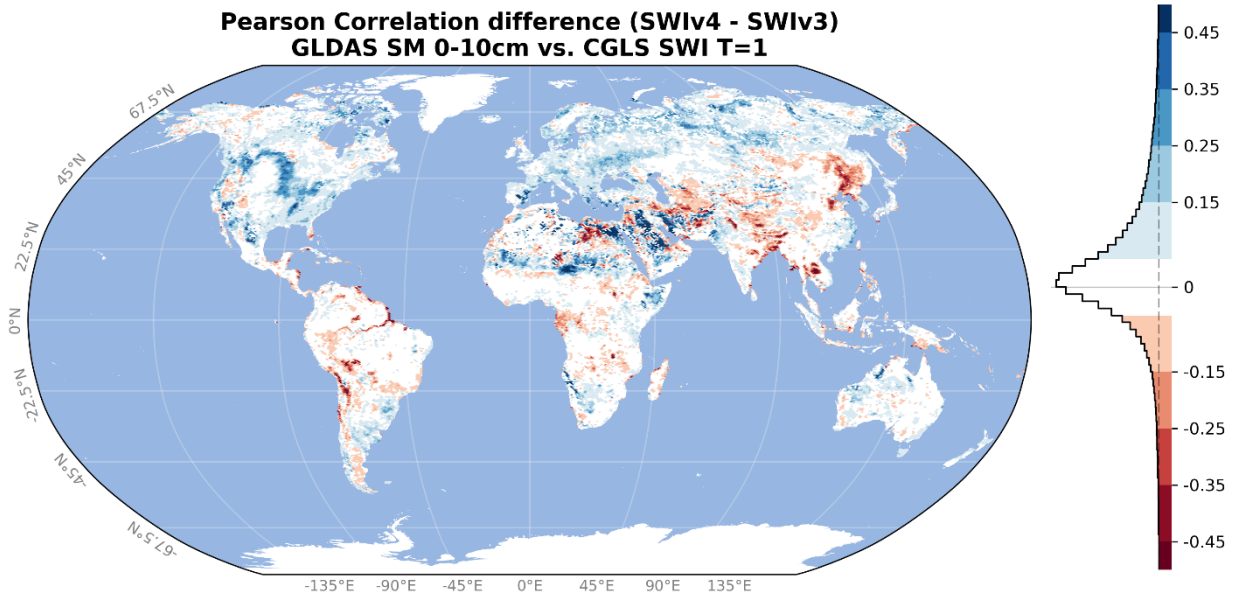


Figure 3: Difference of Pearson correlation (SWIv4 – SWIv3) for T=1 vs. GLDAS NOAH soil moisture 0-0.1m for the comparison validation period (2007-2024)

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4.1.1.2 ubRMSD

Figure 4 and Figure 5 show the obtained ubRMSD values for SWI v3 and SWIv4.0.1 T=1 against GLDAS NOAH soil moisture 0-0.1m and Figure 6 shows the associated difference.

An overall trend to lower ubRMSD values can be observed with biggest differences found in Northern Africa and the Middle East. It must, however, be noted that these areas are mostly dry areas with low soil-moisture dynamics, and also the obtained correlations are largely negative. Therefore, any improvements in ubRMSD in these regions must be viewed with caution since they might well be a result of the more extensive masking rather than an actual improvement in the representativeness of soil-moisture dynamics. However, the improved masking of SSM observations can still have a significant positive effect on the SWI value by avoiding ingestion of irritated SSM observations into the SWI computation.

The improved SSF masking can also partly explain better agreement in northern latitudes for SWIv4.0.1 (with the new adaptive vegetation correction in the HSAF input certainly also contributing to this result).

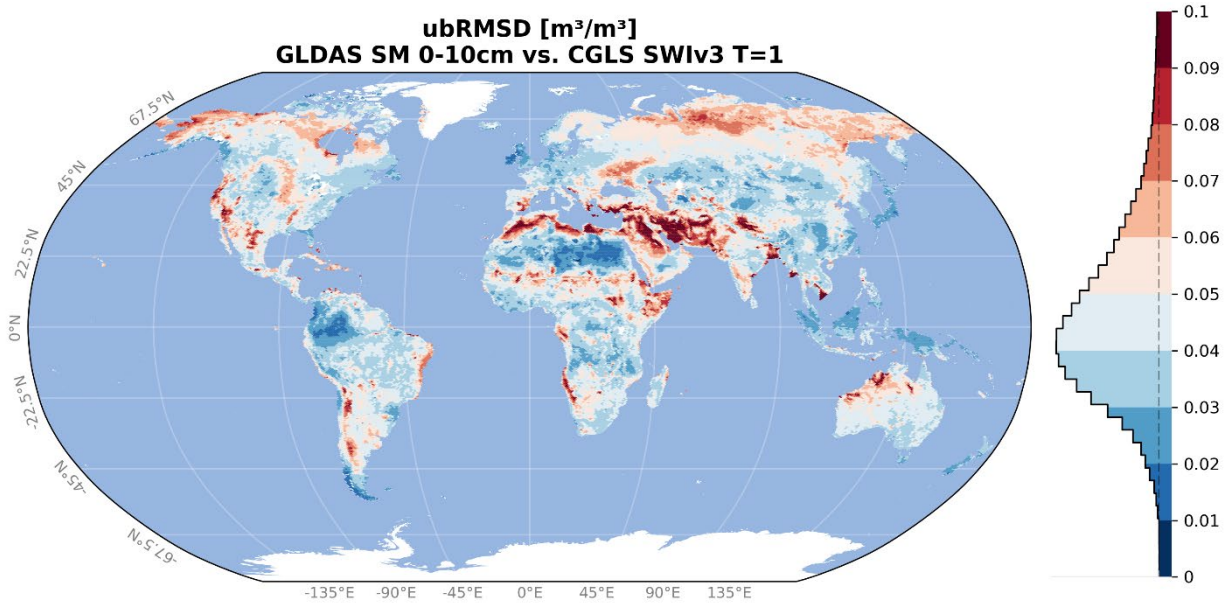


Figure 4: ubRMSD [m³/m³] of SWIv3 T=1 vs. GLDAS NOAH soil moisture 0-0.1m for the comparison validation period (2007-2024)

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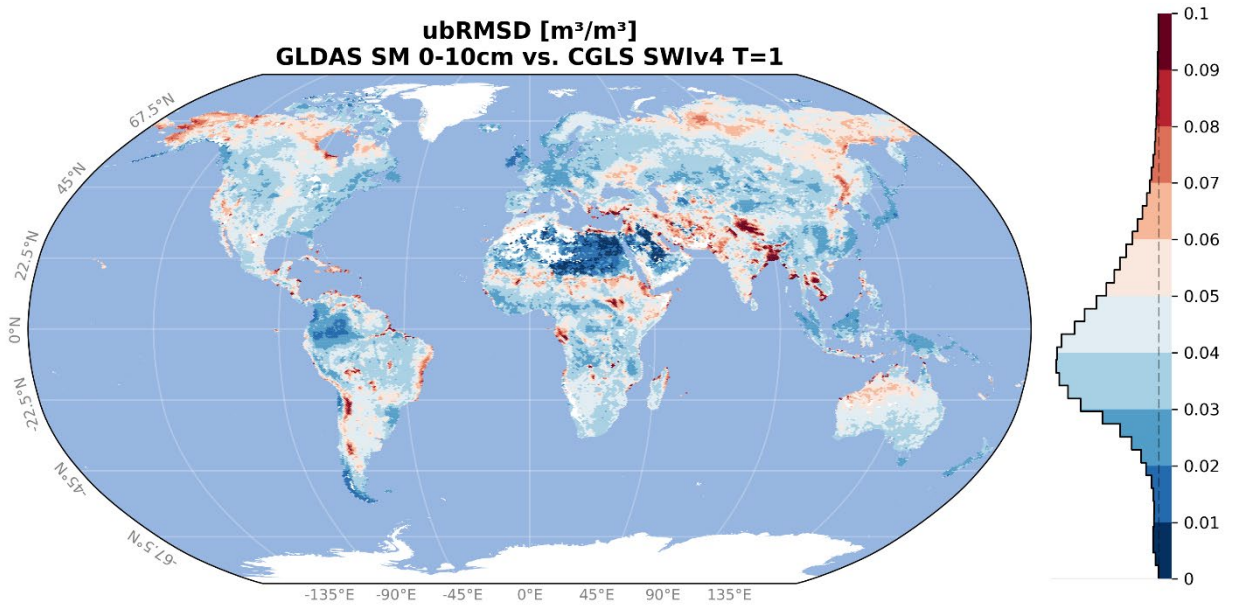


Figure 5: ubRMSD [m³/m³] of SWIv4 T=1 vs. GLDAS NOAH soil moisture 0-0.1m for the comparison validation period (2007-2024)

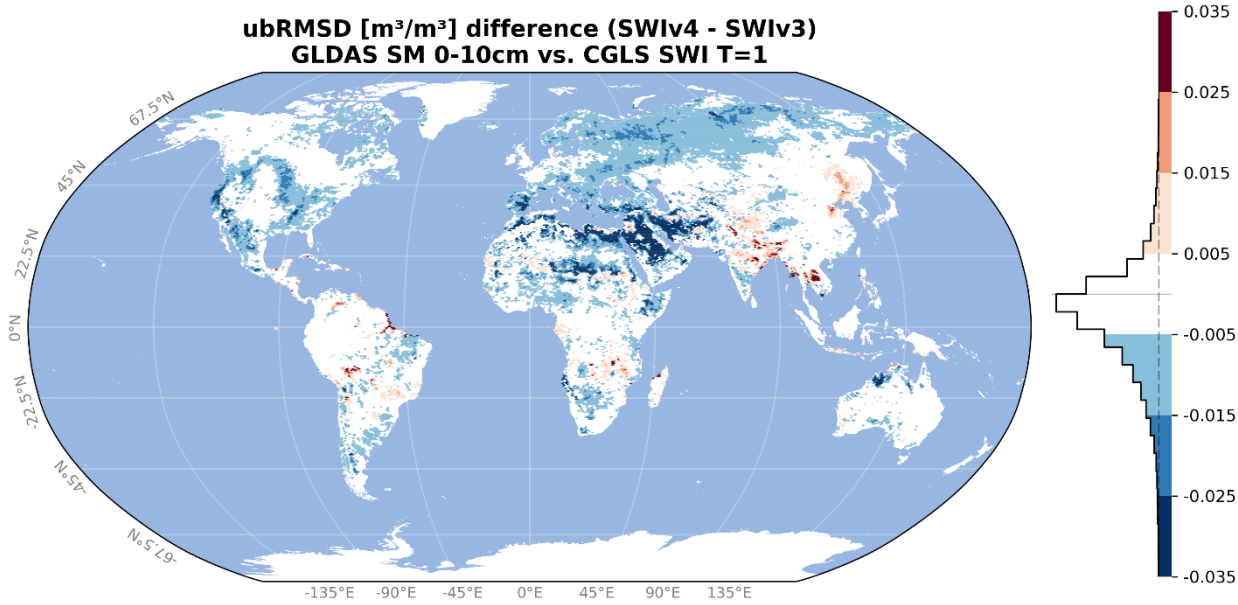


Figure 6: Difference of ubRMSD [m³/m³] (SWIv4 – SWIv3) for T=1 vs. GLDAS NOAH soil moisture 0-0.1m for the comparison validation period (2007-2024)

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4.1.2 Comparison of SWIv3 and SWIv4.0.1 performance metrics for in-situ validation

This section presents a comparison of performance metrics obtained for in-situ validation of SWIv3 and SWIv4.0.1.

4.1.2.1 Pearson Correlation

Figure 7 shows a comparison of the Pearson correlation in-situ validation results for SWIv3 and SWIv4.0.1 for single-year and multi-year periods.

It is clearly visible that SWIv4 provides an improved performance for all presented validation periods.

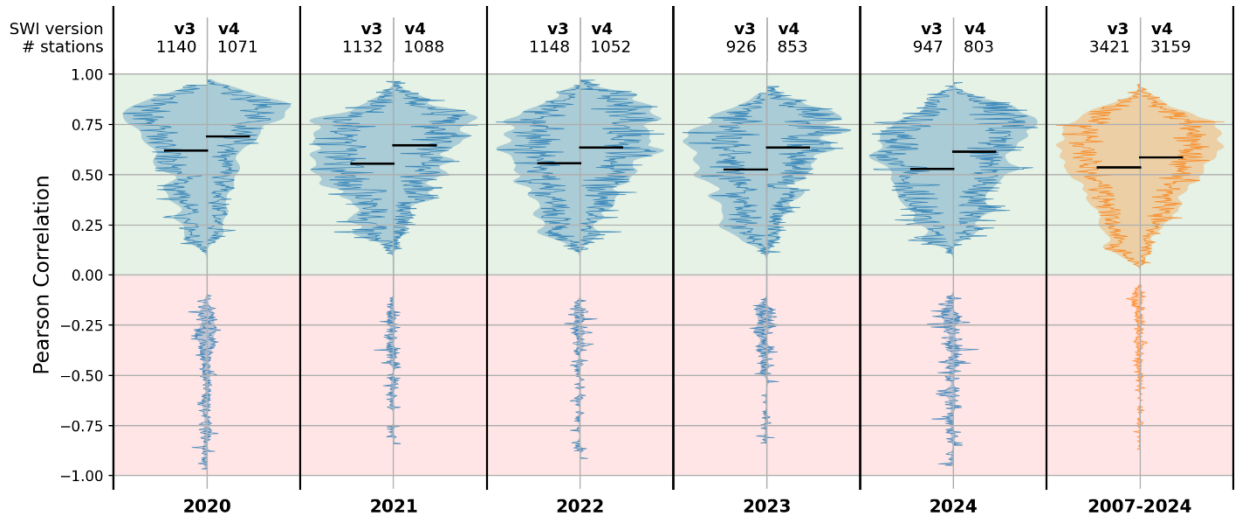


Figure 7: Comparison of yearly and multi-year Pearson correlation distributions for SWIv3 T=1 and SWIv4.0.1 T=1 against all available in-situ stations (depth < 0.1m) of the ISMN network. The black line indicates the median of the distribution. The corresponding SWI version and the number of stations that fulfil the criteria defined in Section 3.4 are indicated above the distributions.

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4.1.2.2 ubRMSD

Figure 8 shows a comparison of the ubRMSD in-situ validation results for SWIv3 and SWIv4.0.1 for single-year and multi-year periods.

Similar to Figure 7, a lower ubRMSD is obtained for SWIv4 for all presented validation periods.

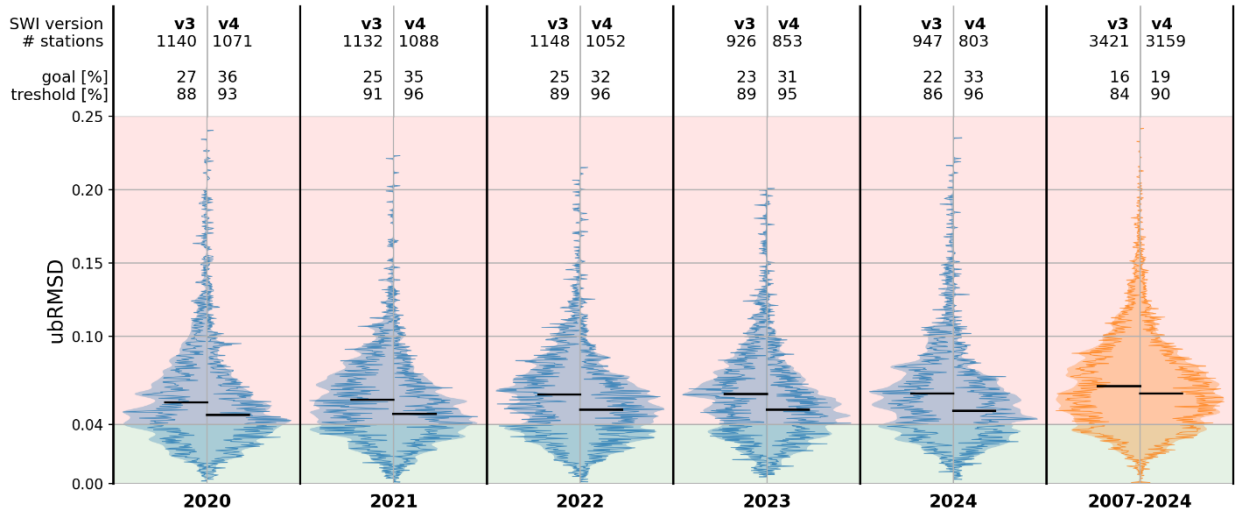


Figure 8: Comparison of yearly and multi-year ubRMSD distributions for SWIv3 T=1 and SWIv4.0.1 T=1 against all available in-situ stations (depth < 0.1m) of the ISMN network. The black line indicates the median of the distribution. The corresponding SWI version and the number of stations that fulfil the criteria defined in Section 3.4 are indicated above the distributions. Light-green shading indicates the region where the goal requirements are met. Moreover, the percentage of cases satisfying threshold and goal requirements, respectively, are given in the heading of the graphics.

4.1.3 Inter-Comparison of SWIv3 and SWIv4

This section presents an inter-comparison of SWIv3 and SWIv4.0.1, highlighting expected differences between the major product versions.

4.1.3.1 Pearson Correlation

Figure 9 shows the obtained Pearson correlation between SWIv3 and SWIv4.0.1 for T=1.

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The overall very high correlation values indicate that the value dynamics of the datasets remain very similar with most prominent differences in dry areas such as the Sahara and large parts of the Middle East.

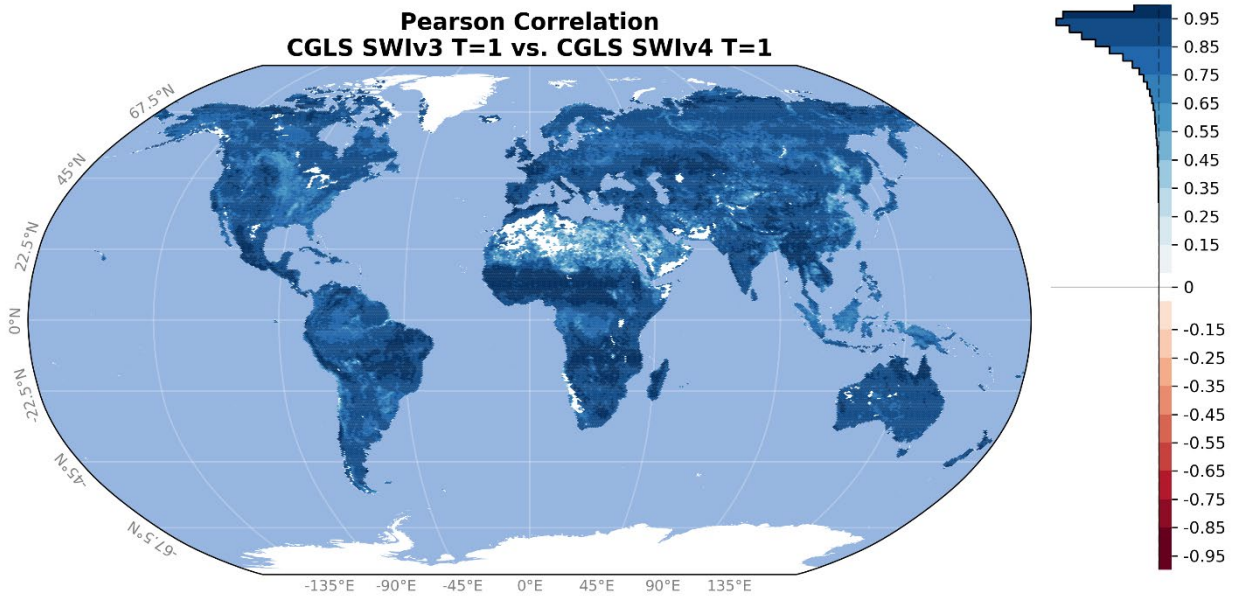


Figure 9: Pearson correlation of SWIv3 T=1 vs. SWIv4 T=1 for the comparison validation period (2007-2024)

4.1.3.2 ubRMSD

Figure 10 shows ubRMSD between SWIv3 and SWIv4 for T=1 in units of % relative saturation.

In most areas, ubRMSD between the dataset versions is below 10%. The biggest differences occur in densely vegetated areas such as the Amazon and Congo rain forests and South- to South East Asia, demonstrating the positive effects of the enhanced adaptive vegetation correction incorporated in the new H SAF SSM input data stream.

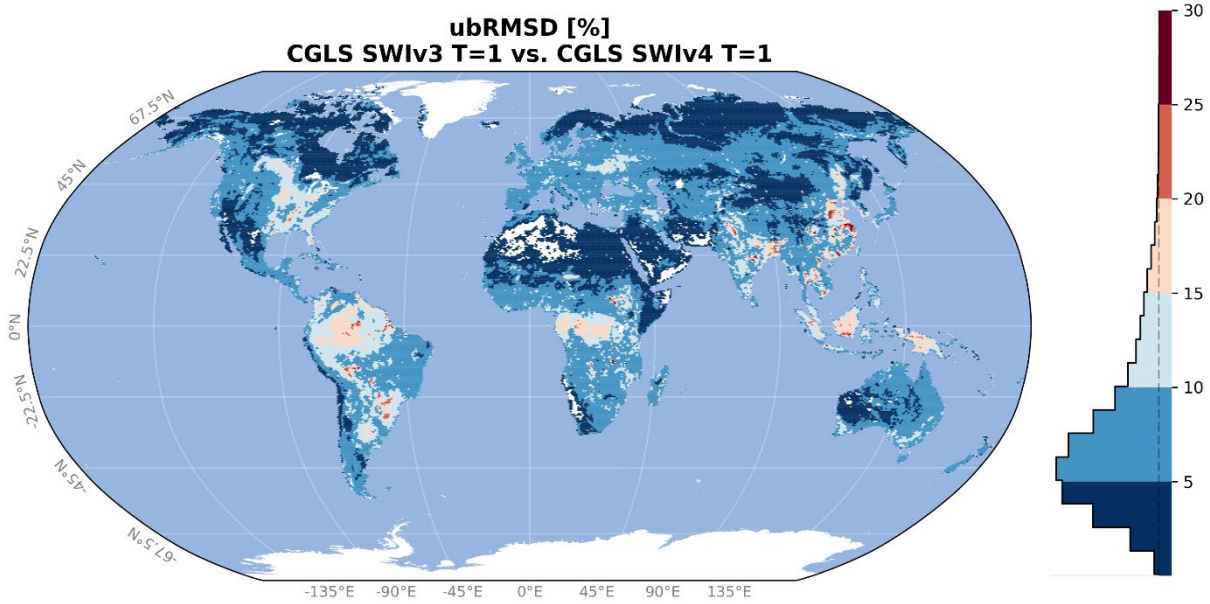


Figure 10: ubRMSD [%] of SWIv3 T=1 vs. SWIv4 T=1 for the comparison validation period (2007-2024)

4.1.3.3 Bias

Figure 11 shows the bias (i.e. difference of mean value) between SWIv3 and SWIv4 for T=1 in units of % relative saturation.

The observed bias-distribution results from the changes in the input data stream directly propagated into the new SWIv4 dataset (cf. section 3.1). Particularly, the frozen surface conditions provided by the new H SAF H29 SSM product used in SWIv4 instead of the previous SSF to mask potentially irritated SSM observations has a drastic effect in regions with pronounced cold seasons (note the pronounced drying of the northern latitudes in SWIv4 in Figure 11; Figure 6 confirms that this is a change for the better). Conversely, the other augmentations introduced in H29 SSM result in increased SWI values in wide areas around the Red Sea and the southern United States, for example.

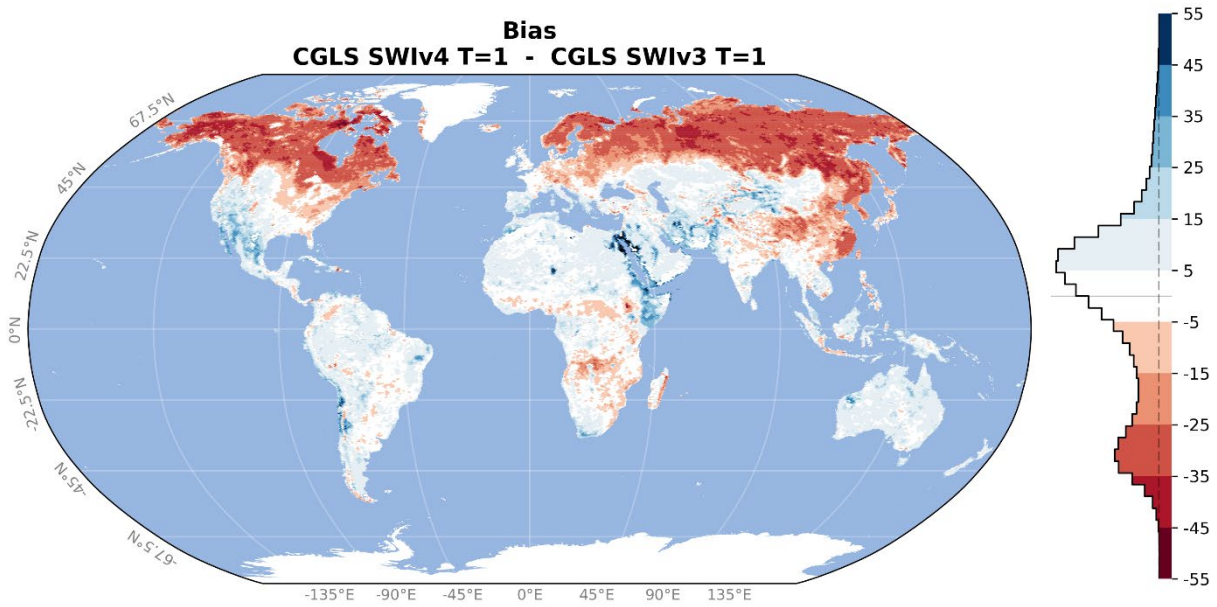


Figure 11: Bias SWIv4 T=1 minus SWIv3 T=1 for the comparison validation period (2007-2024)

4.1.3.4 Temporal Masking

A major difference between SWIv3 and SWIv4 is the implementation of temporal masking with respect to the associated Quality Flags (QFLAG).

While in SWIv3, no masking based on QFLAG was performed, irrespective of its value (effectively propagating constant SWI values in time until a new soil moisture measurement is available), SWIv4 now explicitly masks any SWI values for which the QFLAG is below 5%.

This results in a much higher degree of masking in mountainous areas, deserts and densely forested regions, as can be seen in Figure 12, Figure 13 and Figure 14 (these figures refer to the entire masking, i.e. including the one based on the SSF).

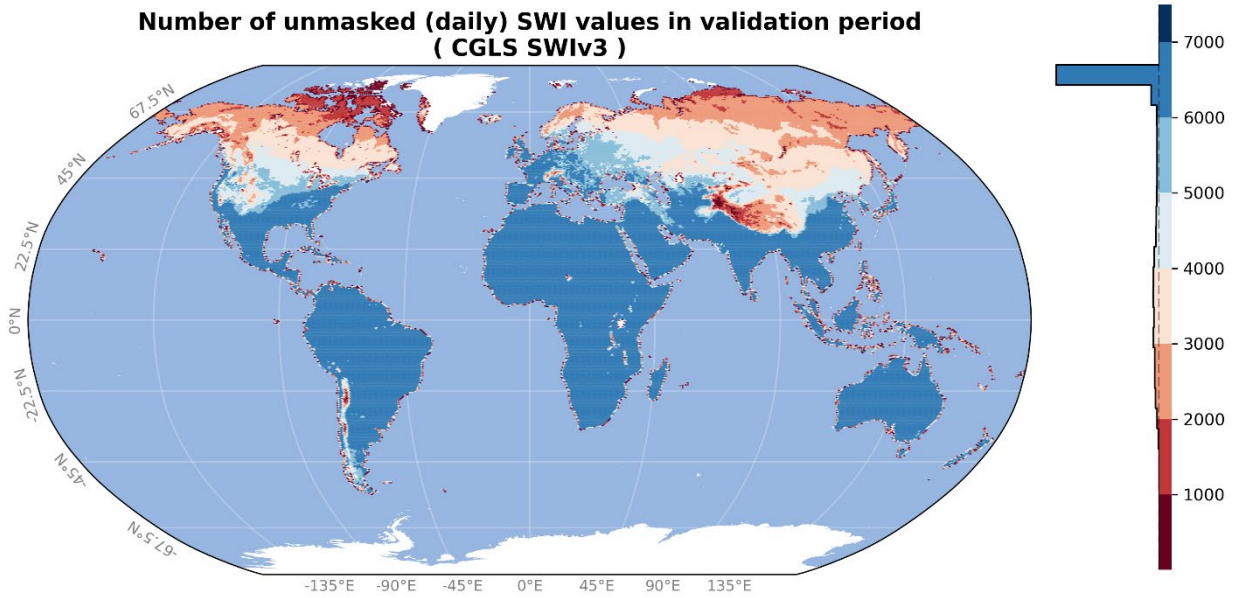


Figure 12: Number of days with unmasked values for SWIv3 T=1 in the comparison validation period (2007-2024)

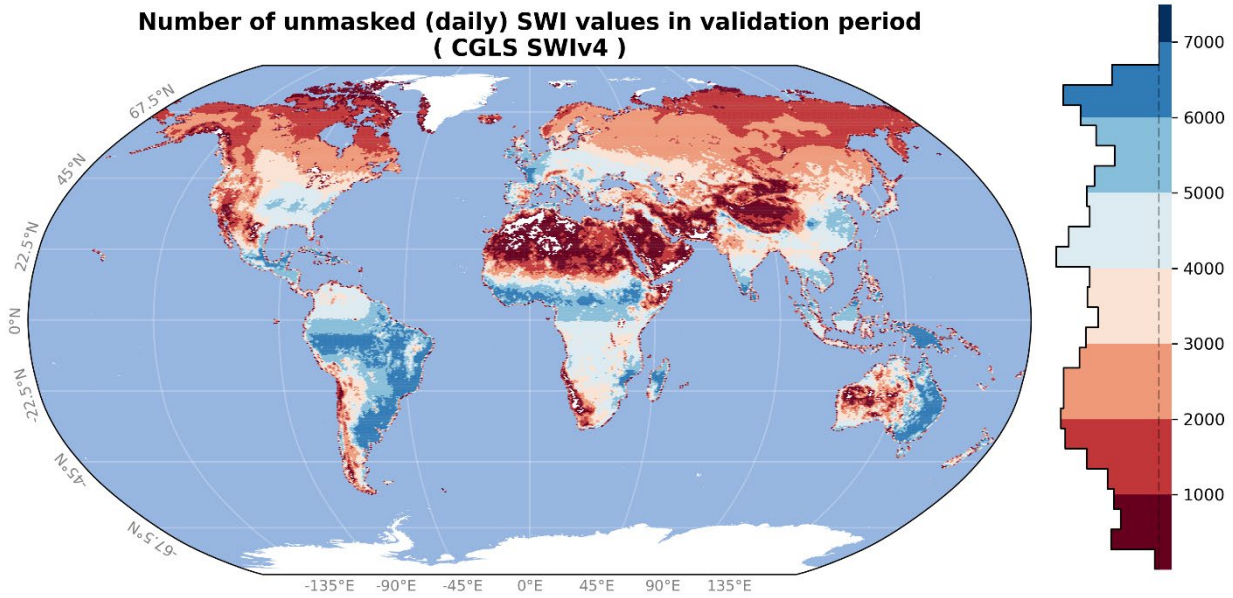


Figure 13: Number of days with unmasked values for SWIv4 T=1 in the comparison validation period (2007-2024)

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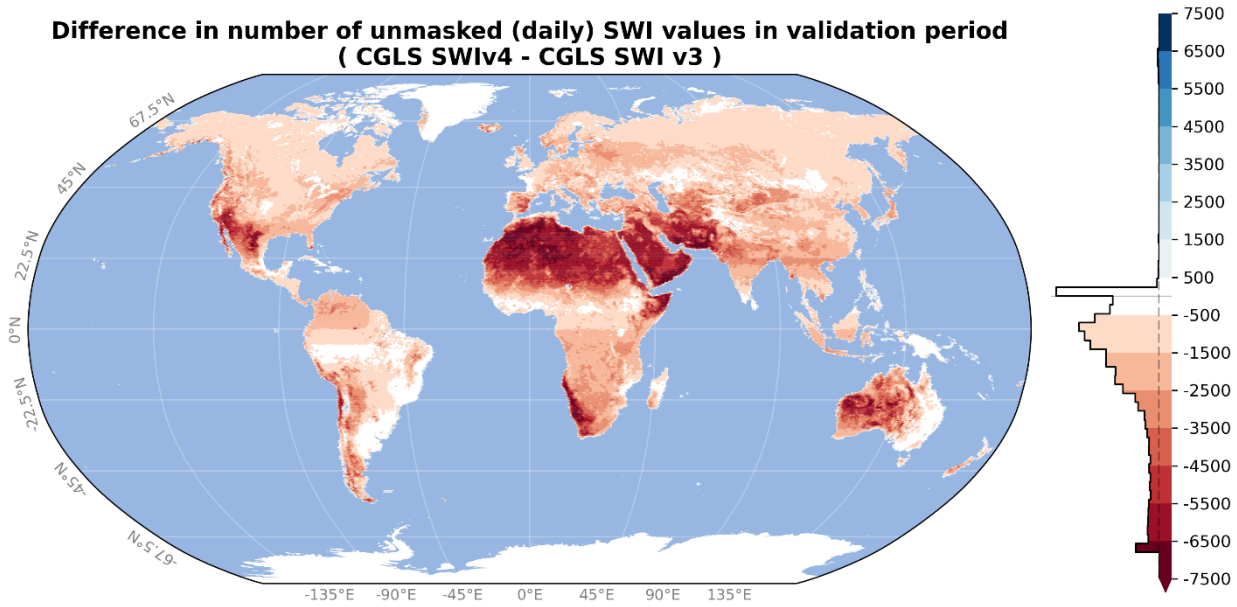


Figure 14: Difference in number of days with unmasked values for SWIv4 - SWIv3 T=1 in the comparison validation period (2007-2024)

4.1.3.5 Effects of temporal masking on obtained performance metrics

A comparison of the performance metrics against the number of available datapoints shows that performance metrics tend to be lower if fewer datapoints are available in the comparison time series (i.e. in areas that require extensive temporal masking).

The reason for this is mostly twofold:

- In case large parts of the seasonal cycle are masked out, the resulting time series will show reduced long-term value dynamics. This will impact scale-invariant performance metrics - such as rho - due to the reduced data variance and also scale-dependent metrics - such as ubRMSD - due to the reduced integrity of the “measurement unit scaling” that is required to bring the time series to the same units.
- Areas that are affected by extensive temporal masking are, in most cases, areas that exhibit difficult soil-moisture retrieval conditions (high vegetation density, high topographic complexity, subsurface scattering). Therefore, it is expected that these areas have lower overall performance scores irrespective of the length of the comparison time series.

In Figure 15 - Figure 17, “Stripes” (high standard deviations of rho/ubRMSD at similar number of observations) are clearly visible at abscissa values of, for example, 4450, 4900, 5300 and

5700 datapoints. These "stripes" originate from the fact that the associated spatial distribution of values represents

1) "latitude-slices" through the Amazon and Congo rainforests and Indonesia, together with
2) a few datapoints from higher temporally sampled regions like central Europe and USA that happen to show the same number of values due to temporal masking (e.g. frozen-soil / snow).

Nevertheless, for number-of-days > 2700, we normally see positive correlations for SWIv3 (Figure 15) and SWIv4 (Figure 16). If temporal masking removes so many points that the remaining time series is even shorter, both statistics rho and ubRMSD deteriorate quickly with smaller number-of-days; the difference plot in Figure 17 suggests that v4 fares somewhat better, particularly in the difficult regions.

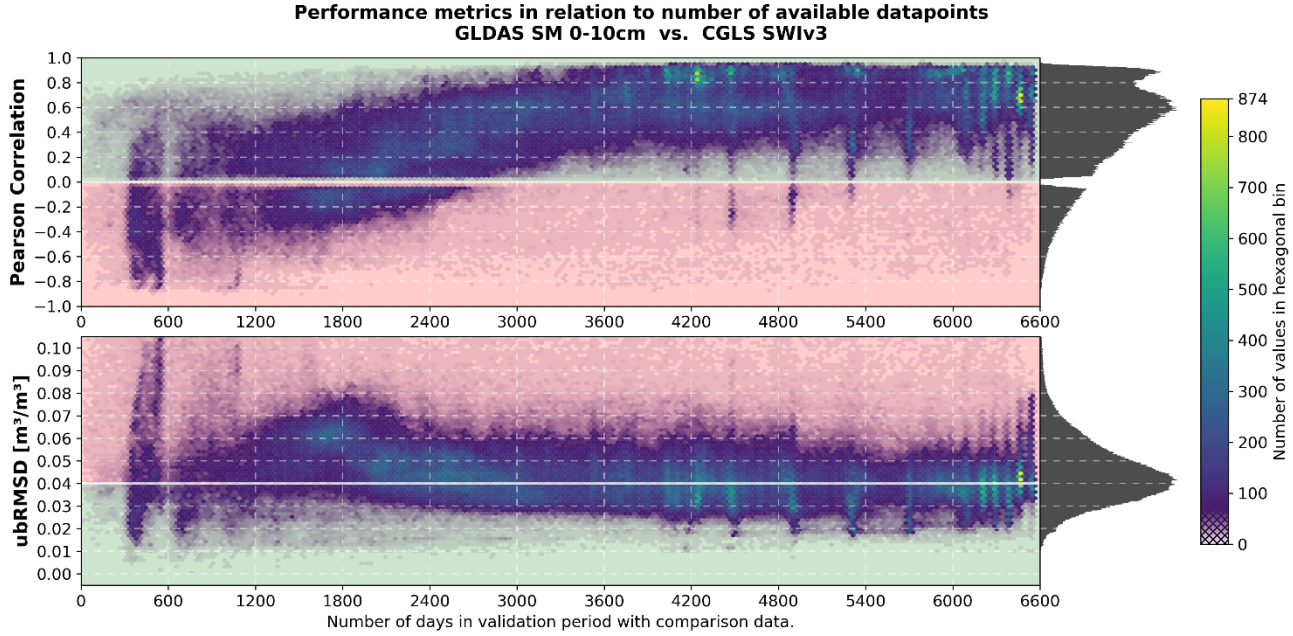


Figure 15: Impact of number of days with comparison data on significant performance metric values for SWIv3 T=1 vs. GLDAS NOAH soil moisture 0-0.1m. Light-green shading indicates the region where the goal requirements are met.

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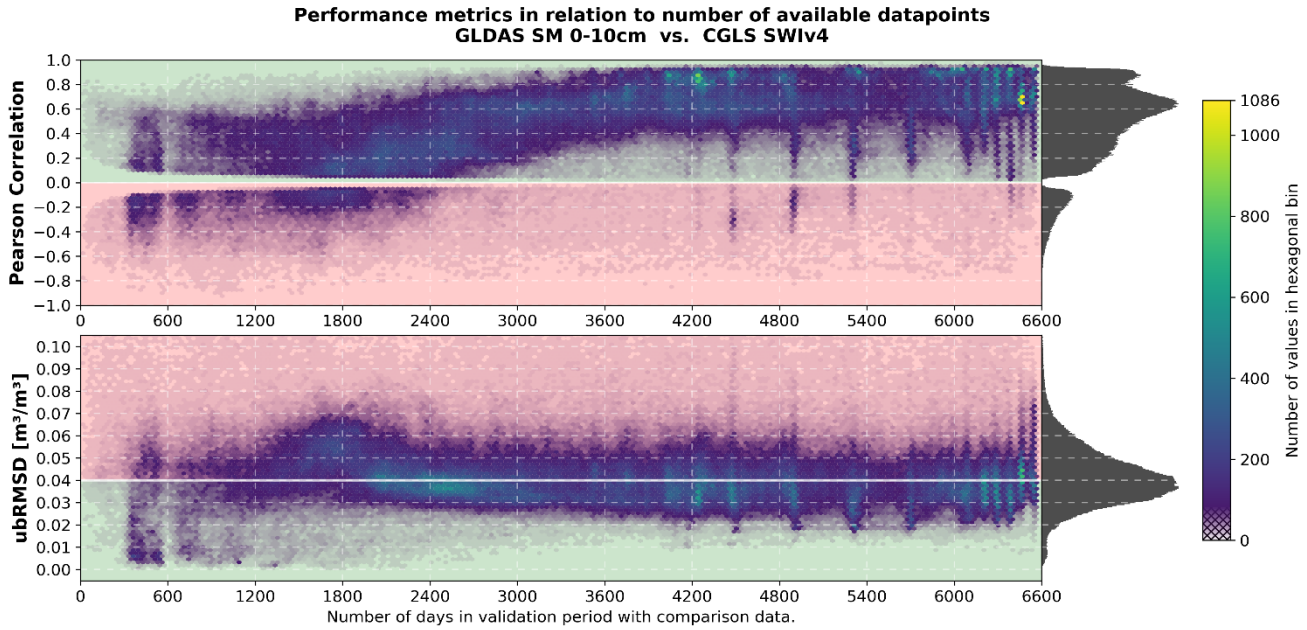


Figure 16: Same as Figure 15, but for SWIv4.

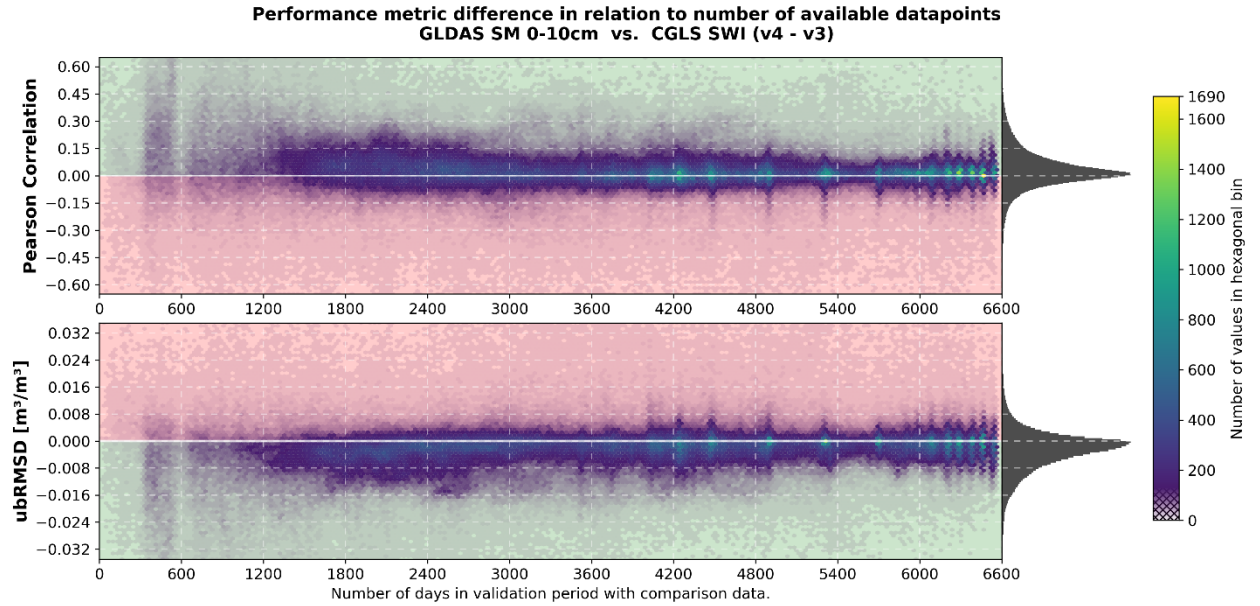


Figure 17: Impact of number of days with comparison data on difference between significant performance metric values for SWIv4 T=1 – SWIv3 T=1 vs. GLDAS NOAH soil moisture 0-0.1m.

4.1.4 Exemplary time series

Three comparisons ISMN vs. SWI v3 vs. SWI v4 in form of time series plots are presented in this section. Only those data points were accepted for the plots where all three sources provided valid values. Consequently, differences shown here cannot be attributed to different masking

strategies, as dates affected by masking were entirely removed (To be exact, masking might well have an influence on such plots for SWI(T=20) or SWI(T=100). However, the latter were generally found to present not much more than smoothed versions of SWI(T=1) time series in the same area. We therefore restrict ourselves to SWI(T=1) plots here).

The first example contrasts the time series at Cochora Ranch, 35.1165°N/119.572°W, California, USA (Figure 18). By and large, the three compared data sets capture the seasonal fluctuations similarly. The correlation coefficients confirm this impression: correlation between SWIv3 and ISMN is 0.80 and rho=0.77 for SWIv4 vs. ISMN. Still, there is an obvious systematic difference in dry seasons where the SWIv3 tends to go distinctly lower, close to the zero-line. During the discussion of Figure 11, the general moistening of the Southern United States was mentioned. This is in agreement with the time series presented here, but its novel aspect is that not all seasons equally contribute to this net effect.

For the time series at La Grand-Combe, 44.2358°N/4.0682°E, France, shown in Figure 19, the differences are not as conspicuous as for the Californian series (France was also an inconspicuous region in the SWIv4-SWIv3 plot of Figure 11). Still there is a spread in the curves in the dry season, at least for some years. Here, however, it is SWI v4 which tends to be below its counterpart in this part of the seasonal cycle. The correlation coefficients indicate that the somewhat larger fluctuations in SWIv4 are closer to the truth as the correlation between SWIv4 and ISMN is 0.75, but merely 0.60 between SWIv3 and ISMN.

It is hardly surprising to find opposite patterns in regions so far apart. The diagrams in fact suggest that dry seasons at Cochora Ranch have a different character than at La Grand-Combe, as the ISMN curve in California often comes very close to 0 whereas the French curve barely falls below $0.05 m^3/m^3$.

Finally, the ISMN station Seolmacheon-B8, 37.9326°N/126.972°E, South Korea, was selected for presentation here, as a representative of several stations in the area where the large difference in correlation coefficients seemed to deserve a closer look. For these series (Figure 20), the correlation between SWIv4 and ISMN was 0.34, which is certainly not too exciting. But the correlation between SWIv3 and ISMN was even clearly negative (-0.18). Thus, it appears in hindsight that SWIv3 computations over that area were essentially useless, yet the introduction of SWIv4 brought some skill which might make it justifiable to use today's CLMS SWI product in this region. One can limit the discussion of the SWIv3 curve by noting that it hardly ever fell below 40% saturation, a pattern not supported by the ISMN curve. Korea was one of the "red" areas in the bias map of Figure 11 as SWIv4 yielded much lower values than SWIv3. The time series in Figure 20 suggests that the recently introduced changes improved the situation by allowing values in the 0%-40% range and this made Korea (numerically) drier.

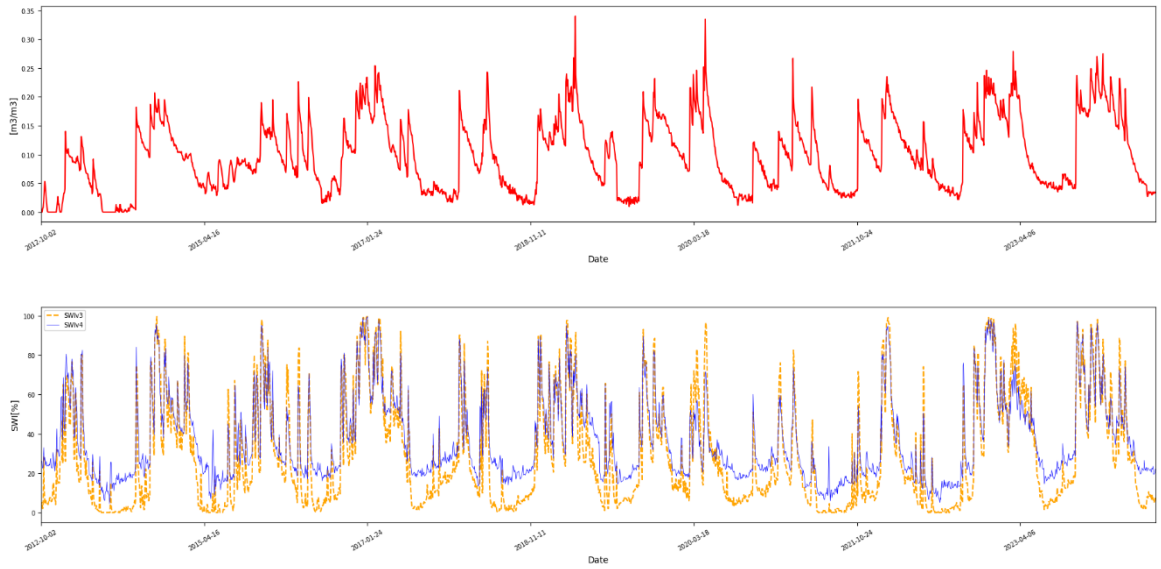


Figure 18: Upper panel: Soil moisture time series (in m^3/m^3) at Cochora Ranch, California. Lower panel: Corresponding time series for SWI v3 (orange) and SWIv4.0.1 (blue), in % saturation, both with T=1.

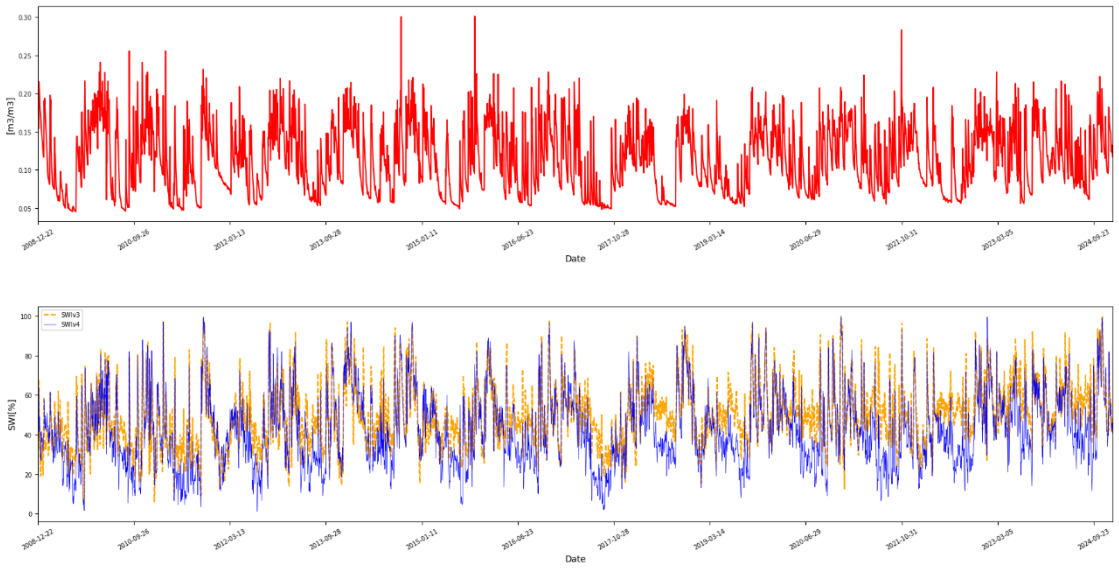


Figure 19: Upper panel: Soil moisture time series (in m^3/m^3) at La Grand-Combe, France. Lower panel: Corresponding time series for SWI v3 (orange) and SWIv4.0.1 (blue), in % saturation, both with T=1.

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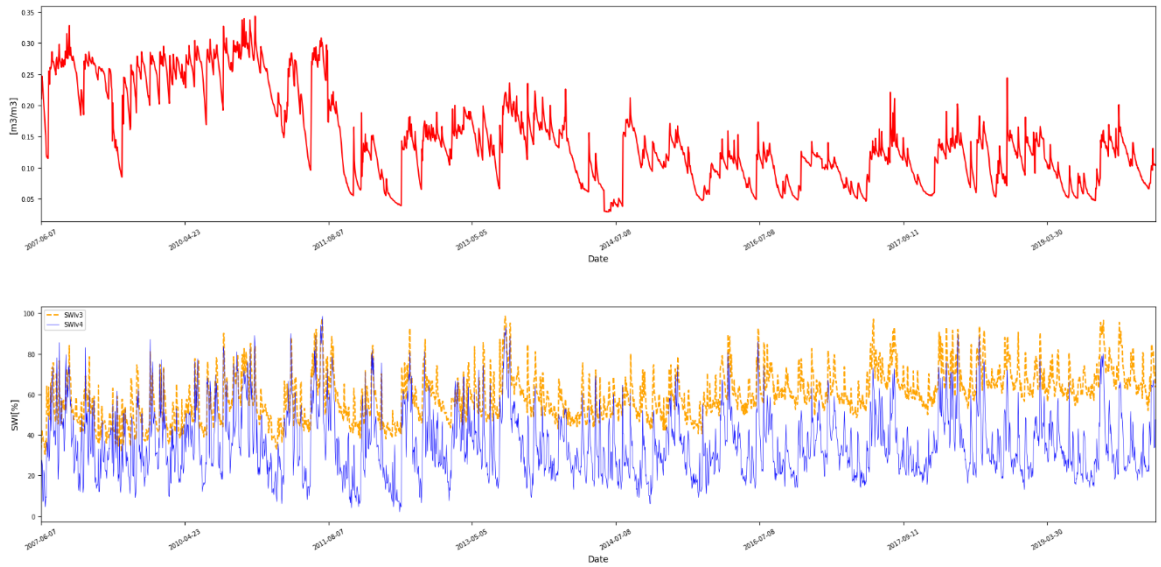


Figure 20: Upper panel: Soil moisture time series (in m^3/m^3) at Seolmacheon-B8, South Korea. Lower panel: Corresponding time series for SWI v3 (orange) and SWIv4.0.1 (blue), in % saturation, both with T=1.

4.1.5 Stability

Stability is the property of an instrument to provide similar measurements when the measured quantity remains constant in time. Analysis of multidecadal long-term stability of satellite data records is a key requirement for the applicability of these products for climate observation. Following previous practice for other variables of the CLMS portfolio, stability is understood as the change in bias (apparent error) over a time period, and the assessment is made via the Mann-Kendall trend test (a statistical standard procedure described e.g. in Helsel et al. (2020)). It is a nonparametric test of the null hypothesis that there is no trend in the data. In case it rejects the null hypothesis, it provides the slope of the so-called Theil-Sen robust line (an analogue to the common regression line) for quantifying the rate-of-change.

The reference against which the SWIv4 bias was determined are the measurements of the ISMN network. The spatial matching of the observations was the same as for the comparisons already shown, and also the “unit matching” was the same as introduced in section 3.2. The covered interval is from 11 April 2007 until 31 December 2025 and all biases calculated at the ISMN stations around the globe were aggregated into daily averages. Series where the correlation between ISMN and SWIv4 was negative were excluded (based on the understanding that there is no relevant information in the bias if the series do not correspond at all) and a second aggregate series was constructed for “fairly well matching” series, arbitrarily defined by rho>0.3. Figure 21 displays the latter. It shows a clear seasonal periodicity; recall that the ISMN stations are unevenly distributed around the globe, which means that in the computation of an

average global bias, a summer-winter effect on the Northern Hemisphere is not compensated by opposite signs on the Southern Hemisphere.

Technically, the Python package pyMannKendall (Hussain and Mahmud, 2019) was used to carry out the statistical test. The package offers several variants of the Mann-Kendall test. Two of them applying corrections for autocorrelated data were chosen here, as this was deemed advisable in light of curves like the one in Figure 21: 1) Hamed and Rao Modified Mann-Kendall test (Hamed and Rao, 1988); 2) Modified Mann-Kendall test using a pre-whitening method proposed by Yue and Wang (2002). Fortunately, the two variants always agreed in their conclusions, which simplifies the further discussion. The default criterion in the used software package, which was retained for our analyses, is to reject the "no-trend" hypothesis if the p-value is below 0.05.

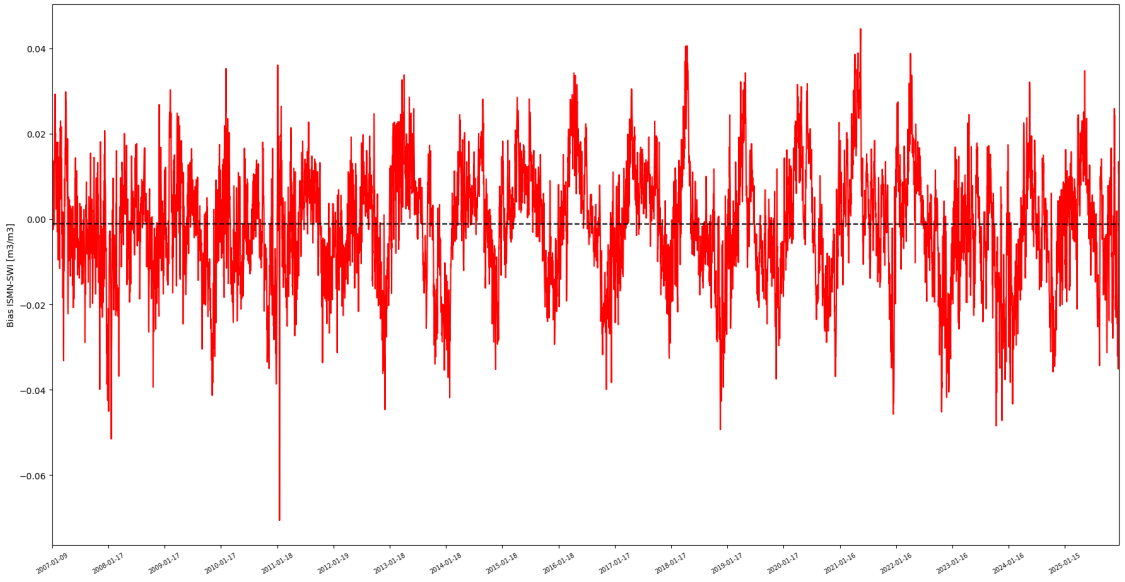


Figure 21: Time series of globally averaged daily biases ISMN – SWIv4 (T=1), 2007-2025, with the robust Theil-Sen line as result of the Mann-Kendall trend analysis superimposed (black, dashed).

As the purpose of the exercise is to gain insight about the stability of the sensor, it is sufficient to test SWI for T=1. The values for T=1 are "closest" to the sensor whereas for longer characteristic times, the influence of the soil infiltration model comes into play. The outcome of the Mann-Kendall tests was "no trend". Numerically, slopes of -0.0017m^3/m^3 or -0.0018 m^3/m^3 (per decade) were obtained (depending on how strict the rho-based filtering was). These figures are mentioned here only as a matter of completeness; following good statistical practice,

there is no proof that the slope is actually unequal to 0, confirming ASCAT's reputation as a very stable sensor. Out of scientific curiosity, the evaluations were carried out for T=20 and T=100 as well. For T=20, another "no-trend" diagnosis was obtained whereas the test for T=100 rejected the null hypothesis and indicated a decreasing trend with -0.0083 m^3/m^3 per decade.

4.2 Annual Verification of SWIv4

This section comprises the annual verification of the operational SWI product version.

A set of performance metrics for SWIv4.0.1 at different T-values against model data provided by GLDAS NOAH and in-situ measurements provided by the ISMN is presented. The matching of T-values and depth-layers of the reference datasets is provided in Table 1.

Metrics obtained for the current period (1.1.2025 – 31.12.2025) are compared against the metrics obtained for the reference period (1.1.2007 – 31.12.2024) to evaluate the stability of the product in the current year against the dynamics of the long-term reference period. Note that this is the first annual verification using SWIv4.0.1, hence there are no sections comparing the results here with those of the precursor annual validation for 2024, as that was based on a different product, SWIv3.

4.2.1 Comparison against GLDAS NOAH

4.2.1.1 Pearson Correlation

Figure 22, Figure 23 and Figure 24 show the obtained Pearson Correlation values for T=1, 20 and 100 days against GLDAS NOAH soil-moisture for the current-period (1.1.2025 – 31.12.2025) and the reference-period (1.1.2007 – 31.12.2024).

Table 3 summarizes the associated global mean and median correlation values.

Table 4 presents the percentiles of absolute differences between correlation values for the current and the reference period, respectively.

Table 3: Mean/median of significant rho values between SWI and GLDAS NOAH.

	Reference period (2007-2024)		Current period (2025)	
	mean	**median**	**mean**	**median**
SWI T=1	0.47	0.53	0.52	0.60
SWI T=20	0.39	0.46	0.43	0.57
SWI T=100	0.38	0.45	0.44	0.58

Table 4: Percentiles of absolute significant differences of rho between SWI and GLDAS NOAH.

	50th percentile	70th percentile	80th percentile
SWI T=1	0.15	0.23	0.29
SWI T=20	0.19	0.3	0.38
SWI T=100	0.25	0.36	0.44

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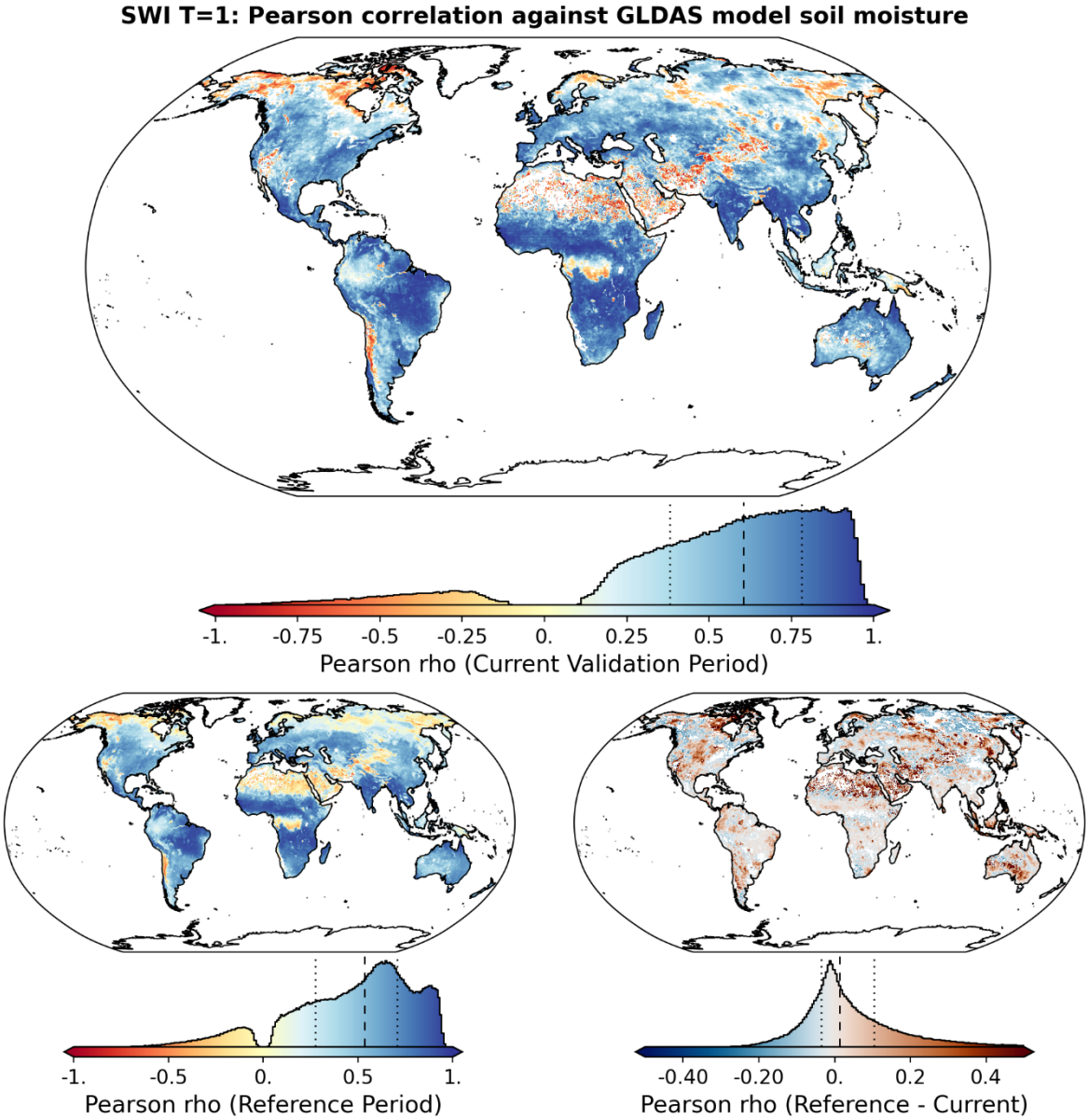


Figure 22: Pearson correlation between SWI T=1 and GLDAS NOAH soil-moisture 0-0.1m for the current period (2025) and the reference period (2007-2024). The histograms show the data distribution as well as median (dashed line) and upper and lower quartile (dotted lines).

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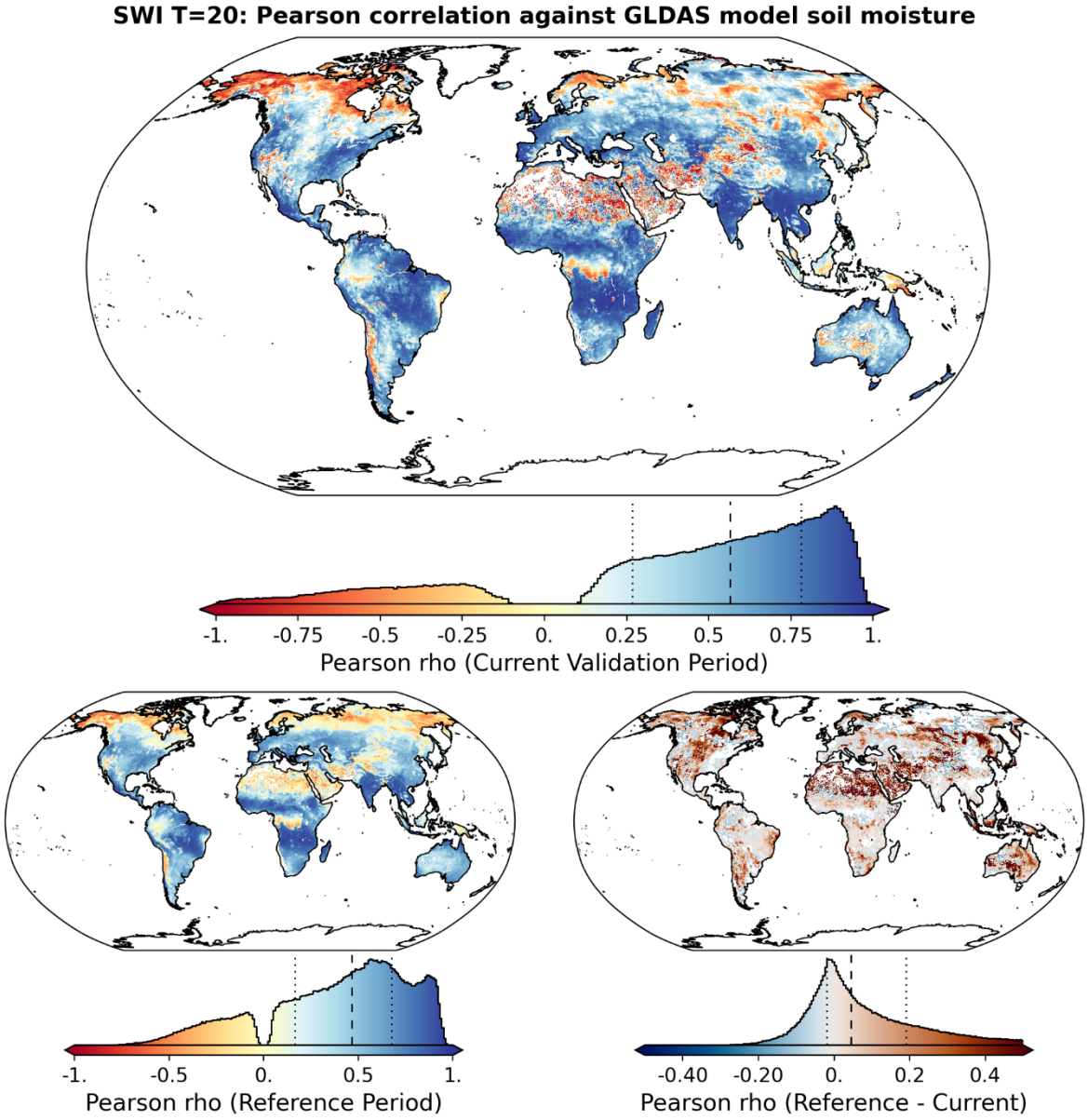


Figure 23: Pearson correlation between SWI T=20 and GLDAS NOAH soil-moisture 0-0.1m for the current period (2025) and the reference period (2007-2024). The histograms show the data distribution as well as median (dashed line) and upper and lower quartile (dotted lines).

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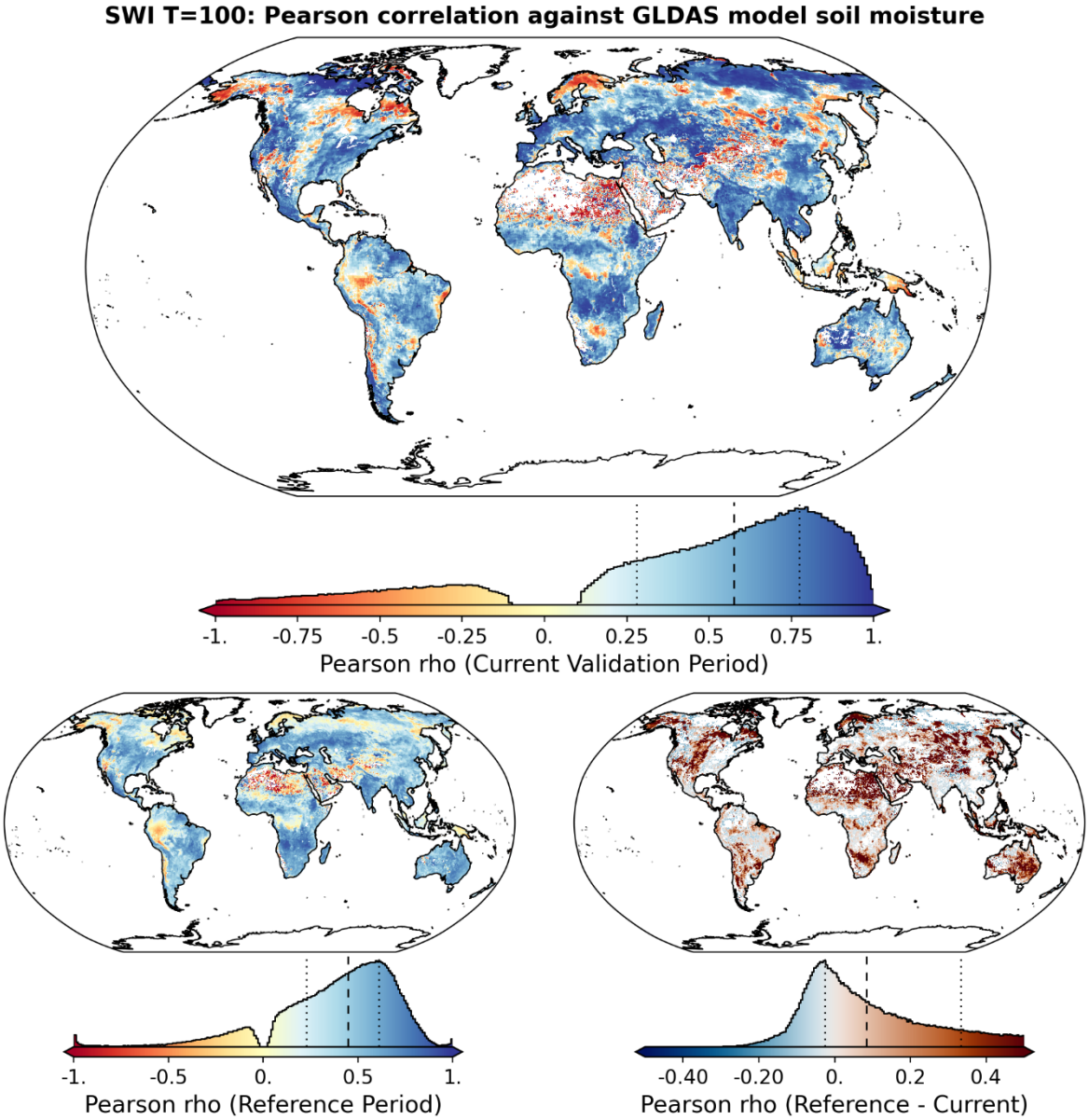


Figure 24: Pearson correlation between SWI T=100 and GLDAS NOAH soil-moisture 0.4-1m for the current period (2025) and the reference period (2007-2024). The histograms show the data distribution as well as median (dashed line) and upper and lower quartile (dotted lines).

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4.2.1.2 ubRMSD

Figure 25, Figure 26 and Figure 27 show the obtained ubRMSD values for T=1, 20 and 100 days against GLDAS NOAH soil moisture for the current period (1.1.2025 – 31.12.2025) and the reference period (1.1.2007 – 31.12.2024).

Table 5 summarizes the associated global mean and median ubRMSD values.

Table 6 presents the percentiles of absolute differences between ubRMSD values for the current period and the reference period.

Table 5: Mean/median of ubRMSD values between SWI and GLDAS NOAH [m³/m³].

	Current period (2025)		Reference period (2007-2024)	
	mean	**median**	**mean**	**median**
SWI T=1	0.05	0.05	0.04	0.04
SWI T=20	0.05	0.05	0.04	0.04
SWI T=100	0.05	0.05	0.04	0.03

Table 6: Percentiles of absolute differences of ubRMSD between SWI and GLDAS NOAH [m³/m³].

	50th percentile	70th percentile	80th percentile
SWI T=1	0.01	0.02	0.02
SWI T=20	0.01	0.02	0.02
SWI T=100	0.01	0.02	0.03

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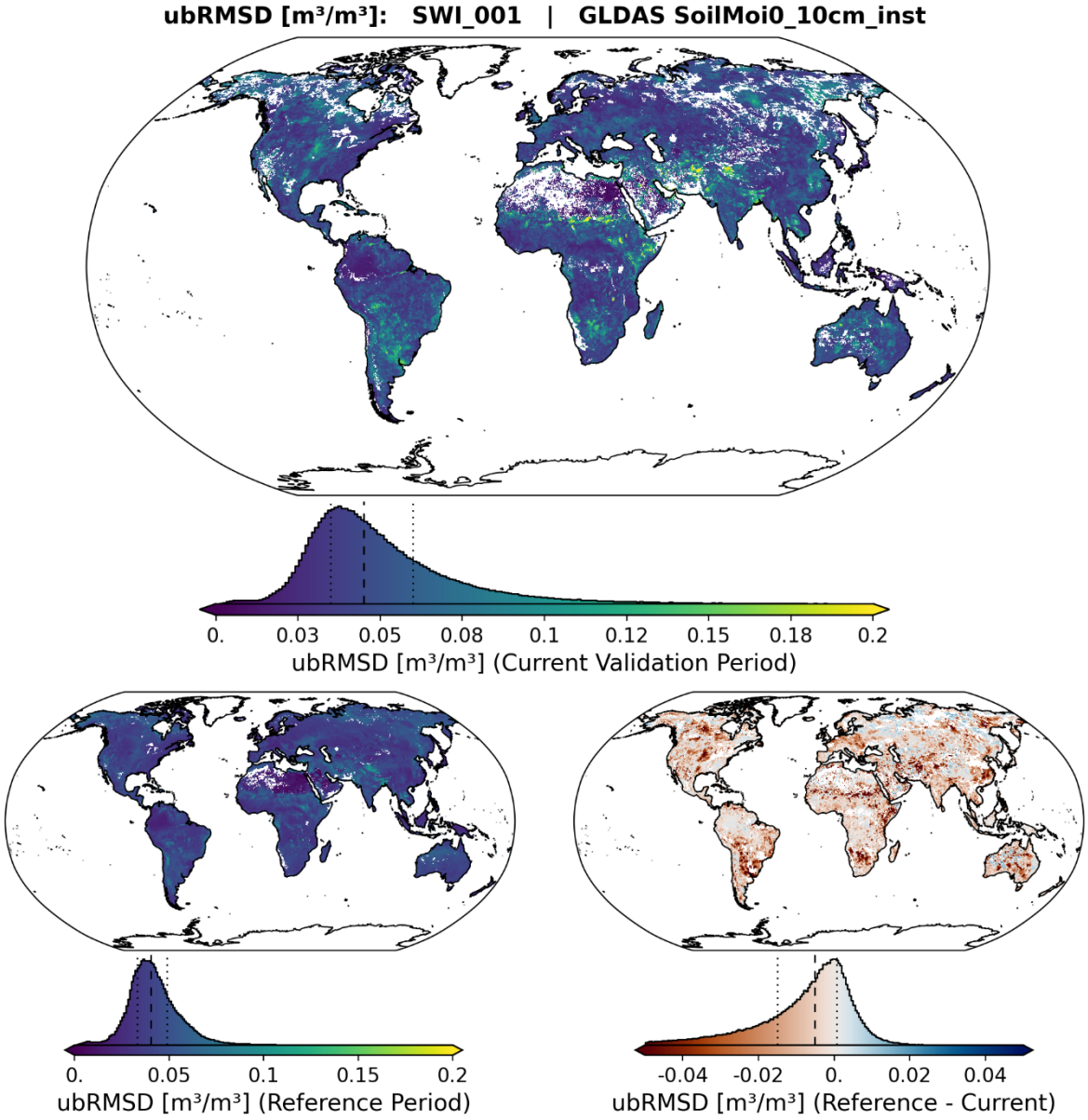


Figure 25: ubRMSD between SWI T=1 and GLDAS NOAH soil-moisture 0-0.1m for the current period (2025) and the reference period (2007-2024). The histograms show the data distribution as well as median (dashed line) and upper and lower quartile (dotted lines).

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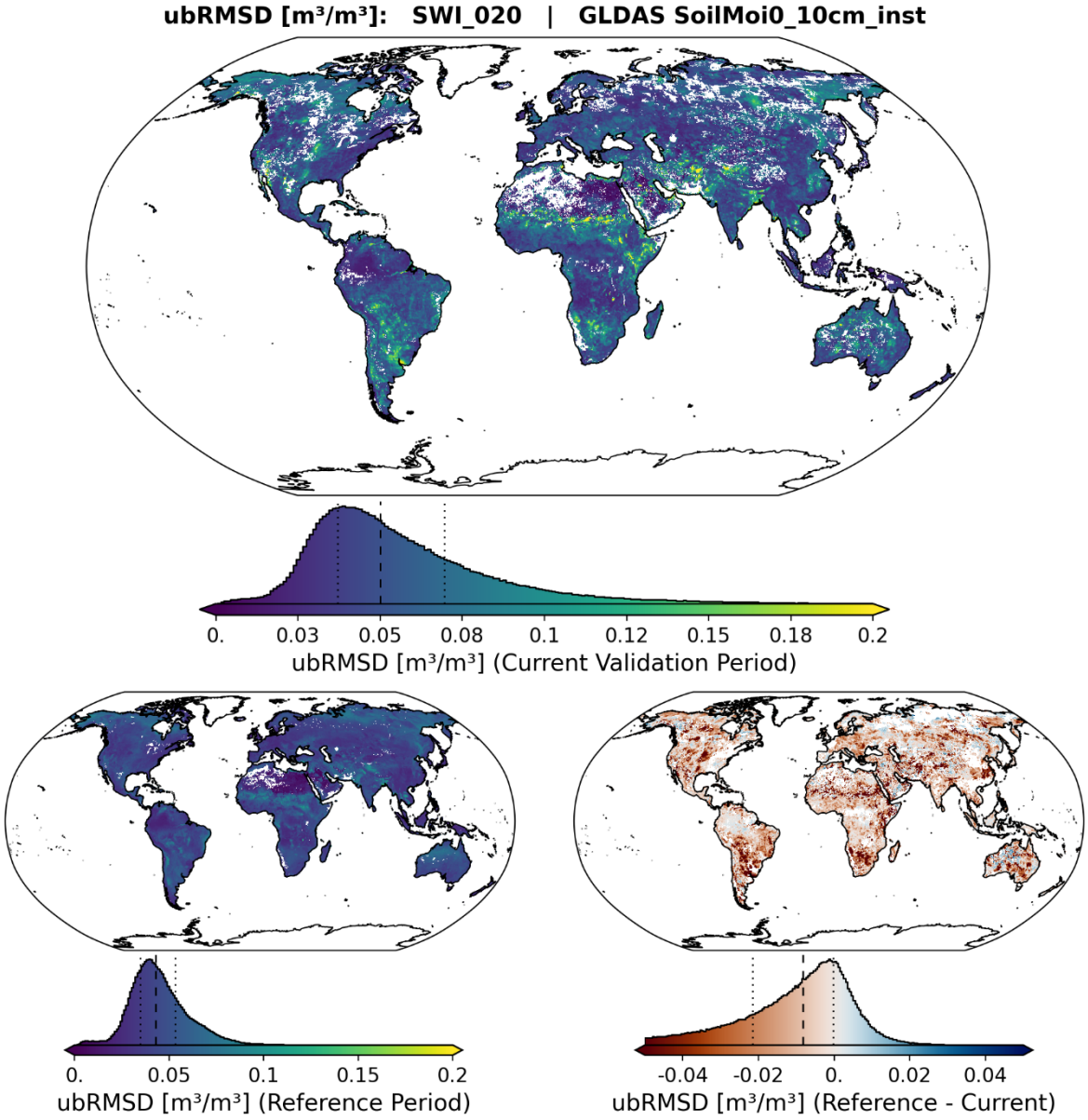


Figure 26: ubRMSD between SWI T=20 and GLDAS NOAH soil-moisture 0-0.1m for the current period (2025) and the reference period (2007-2024). The histograms show the data distribution as well as median (dashed line) and upper and lower quartile (dotted lines).

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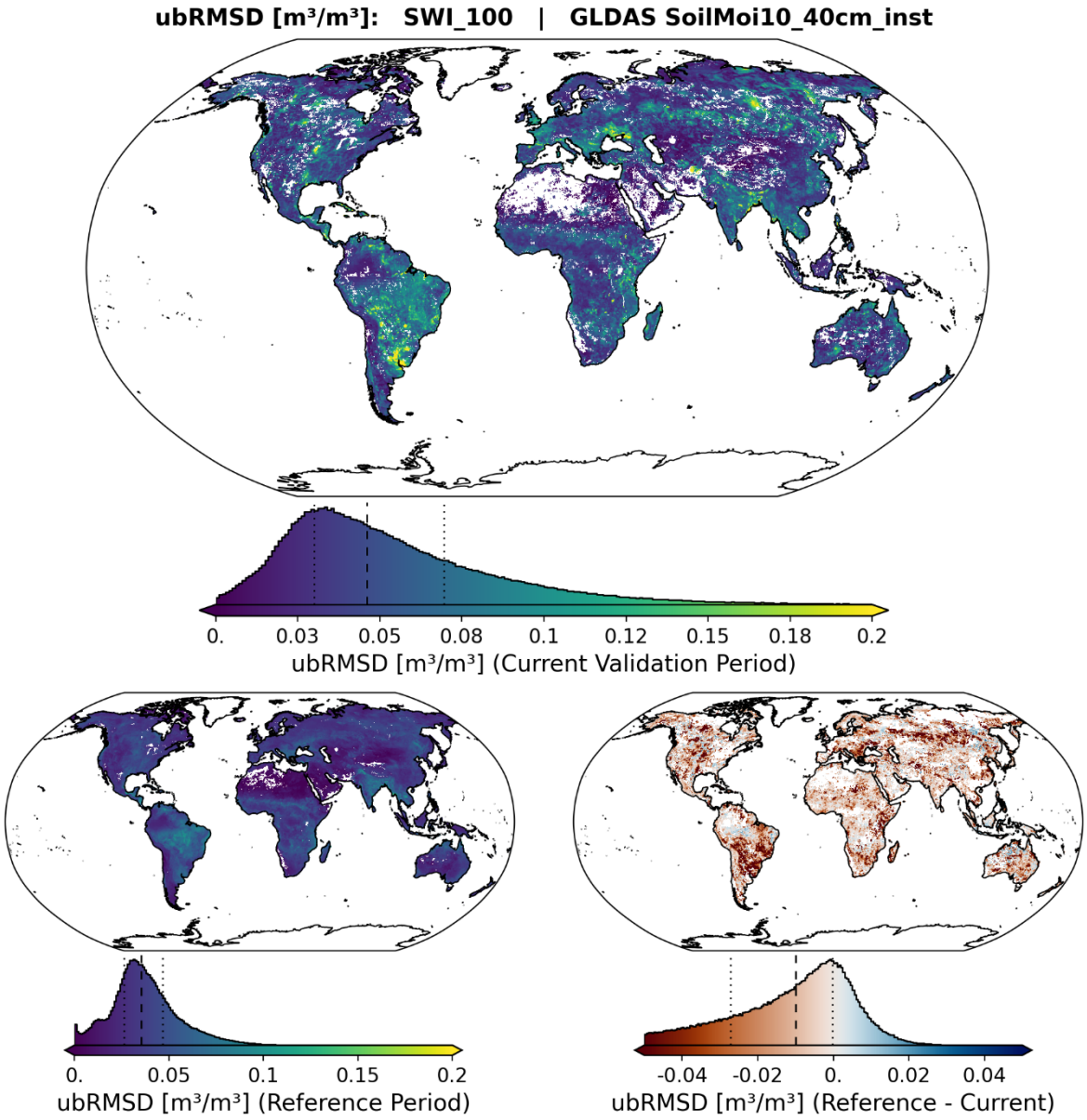


Figure 27: ubRMSD between SWI T=100 and GLDAS NOAH soil-moisture 0.4-1m for the current period (2025) and the reference period (2007-2024). The histograms show the data distribution as well as median (dashed line) and upper and lower quartile (dotted lines).

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4.2.2 Comparison against In-Situ Data

4.2.2.1 General notes on in-situ validation results

It is important to note that the in-situ data validation is affected by both spatial and temporal inhomogeneity. This has the following consequences which must be kept in mind when analysing the obtained performance metric distributions:

- The range of compared years is not the same for each datapoint (see Figure 28);
- The number of timestamps with valid observations for both SWI and in-situ soil moisture has a high variability (from a few hundred to several thousand values) (see Figure 29).

Note: For the continental plots of Figure 28, Figure 29, Figure 30 and Figure 33, the station results are averaged over hexagons with diameter=2.4° degrees, for the sake of readability and to eliminate the visual overlap of plotted stations in areas with dense networks.

The impact of the aforementioned inhomogeneity in the size of the comparison datasets on the obtained performance metrics is illustrated in Figure 31 and Figure 34. A similar analysis with respect to soil moisture data provided by GLDAS NOAH was presented in Section 4.1.3.4.1.

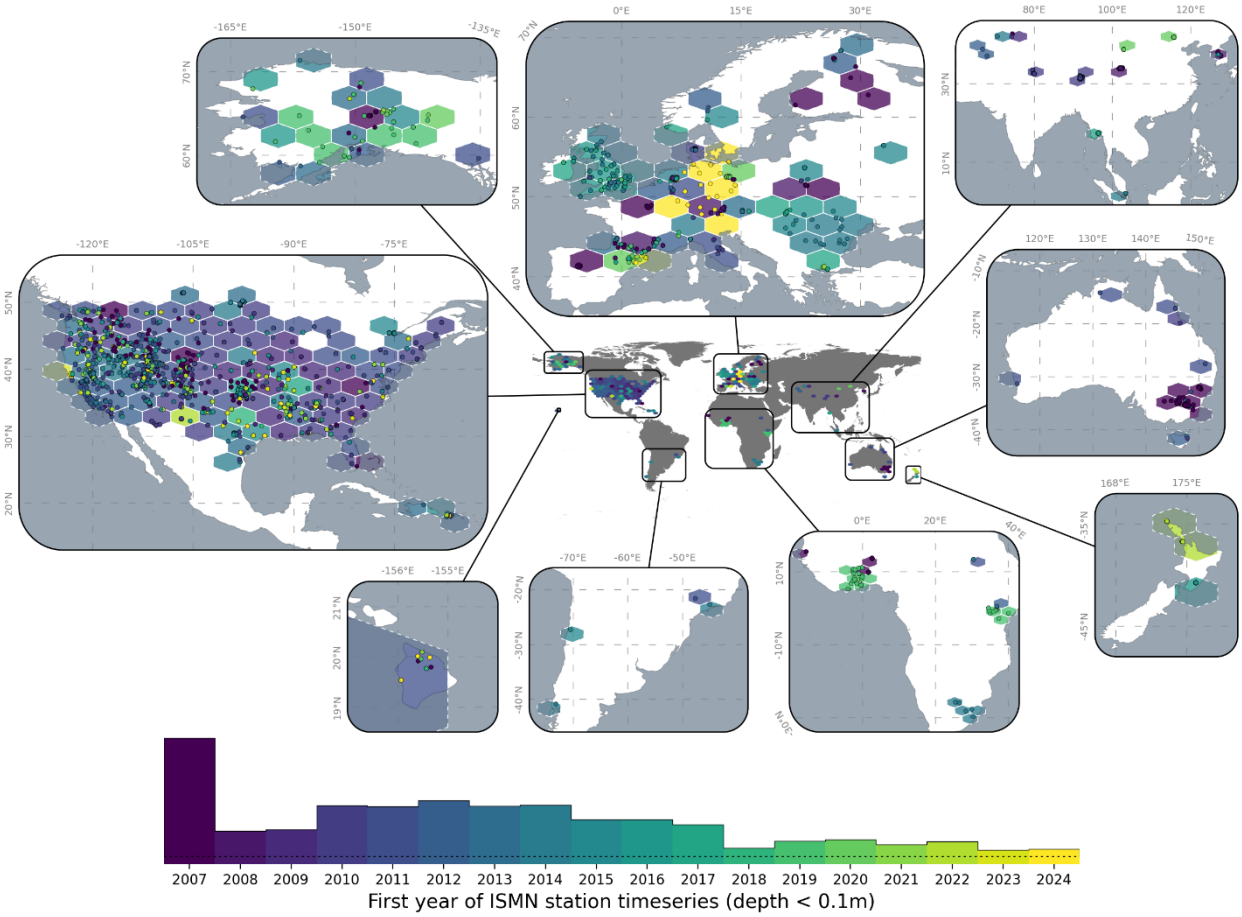


Figure 28: First year in validation period for which ISMN station data is available.

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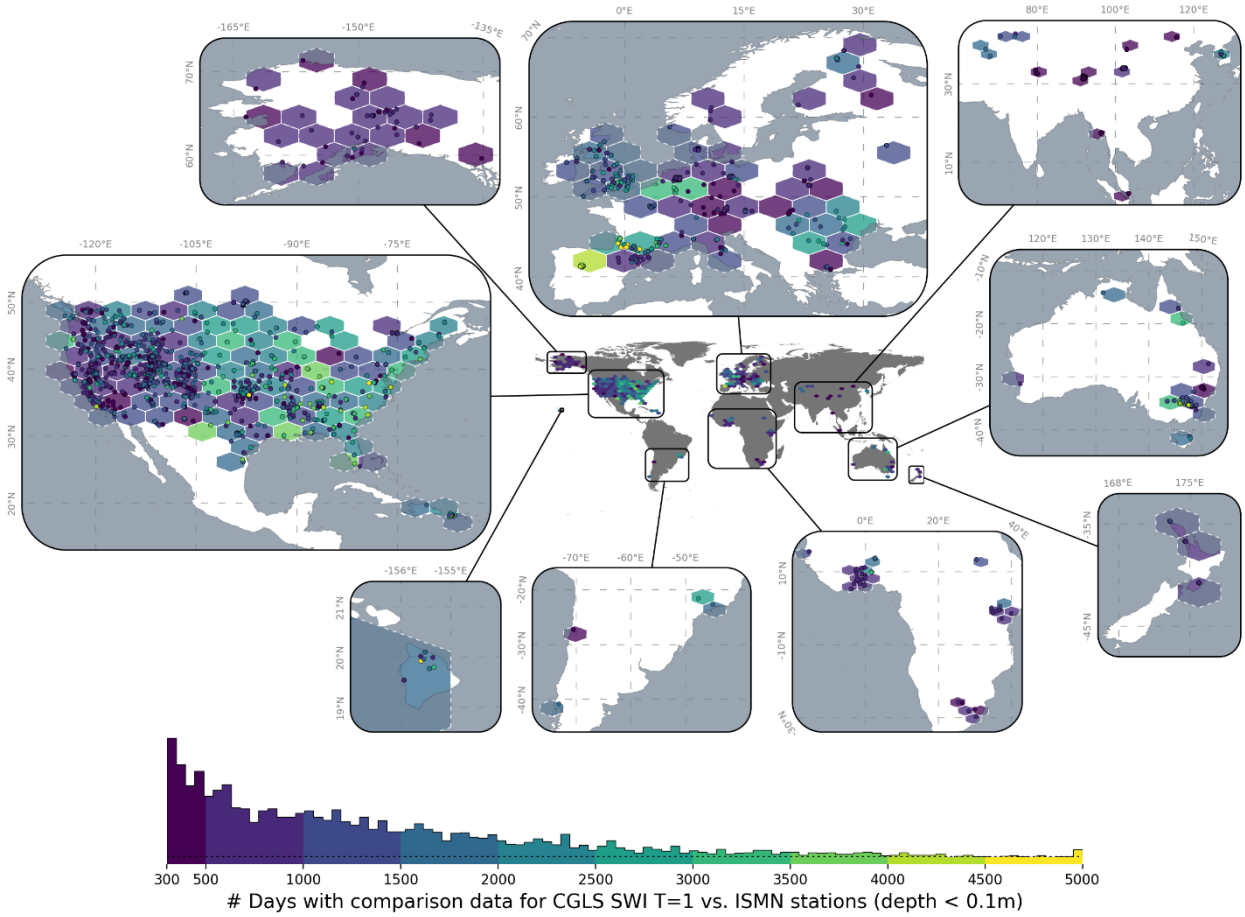


Figure 29: Number of days in validation-period (2007-2025) for which both SWI and in-situ soil-moisture data is available.

4.2.2.2 Pearson Correlation

Figure 30 shows the obtained Pearson correlation between SWIv4.0.1 T=1 and in-situ station data (at a depth <=0.1m). A positive correlation is found for almost all in-situ soil moisture stations with a median correlation of ~0.65, indicating a good agreement between the temporal SWI dynamics and top-layer soil moisture measurements.

Figure 31 illustrates the impact of the comparison-data size on the obtained Pearson correlation. It can be seen that the overwhelming majority of longer time series have high correlations, which highlights the good long-term dynamics of the SWI dataset. Still, the correlations for shorter series are in most cases decent as well, particularly in regions with higher station density. Outliers with negative correlations seem to appear in most cases for small networks or isolated stations.

Figure 32 presents a comparison of single-year Pearson correlations. It can be seen that the SWI continues to provide a stable performance when looking at the last 5 years of data.

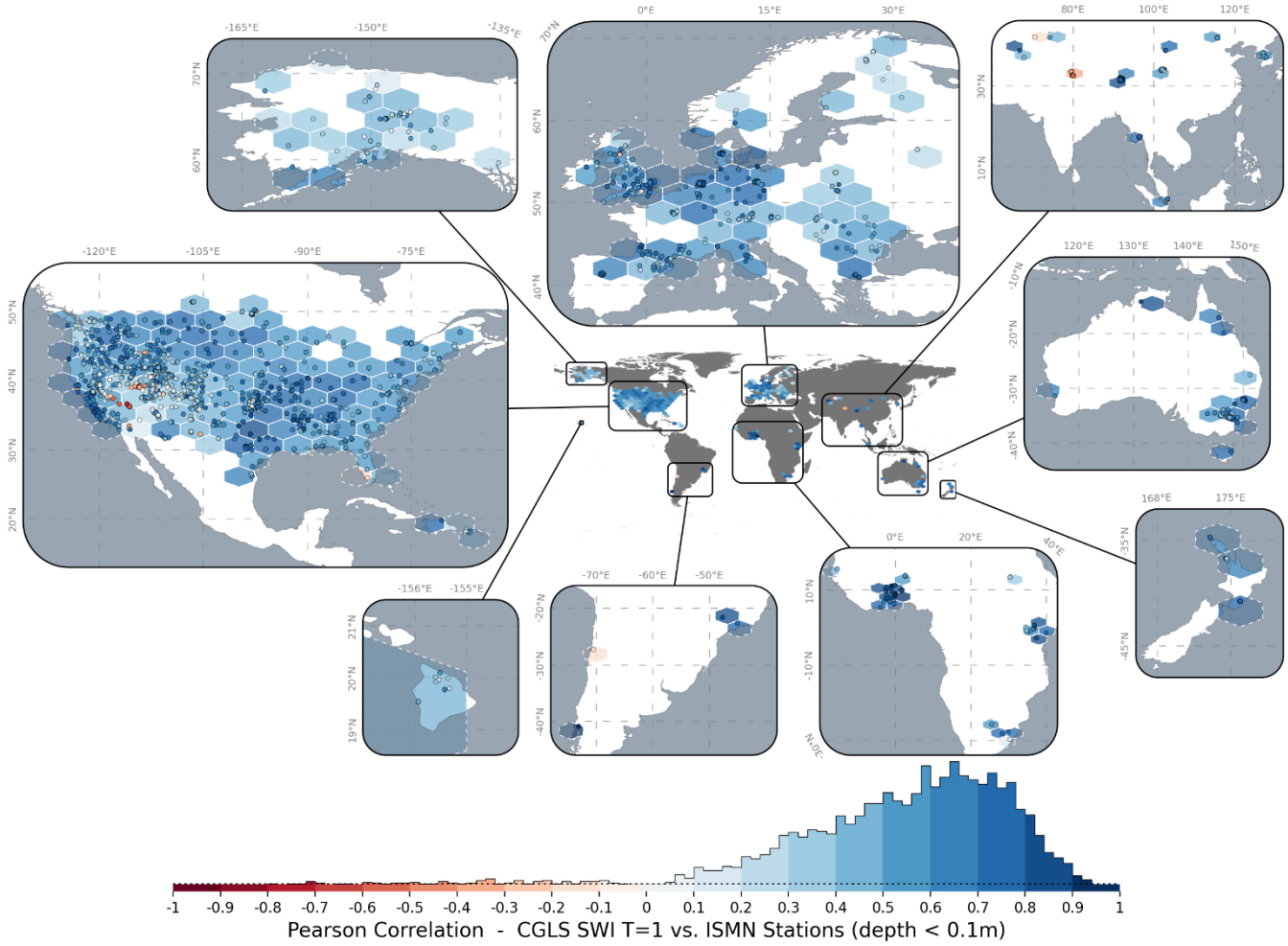


Figure 30: Pearson correlation of CLMS SWI T=1 vs. ISMN station data (depth < 0.1m).

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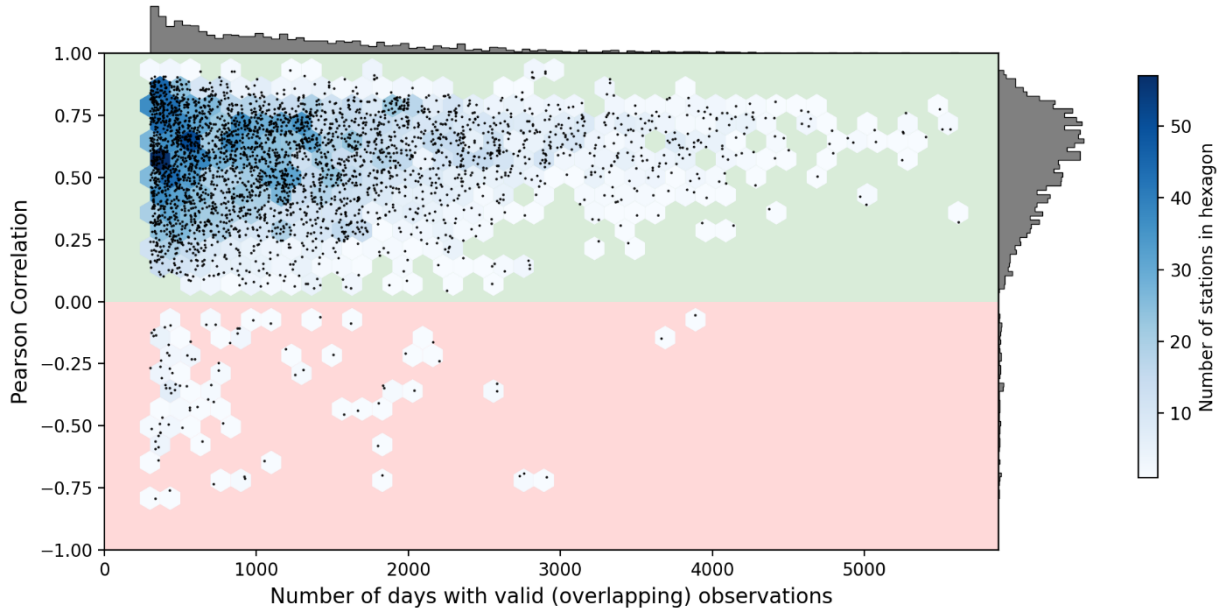


Figure 31: Dependency of Pearson correlation of SWI T=1 vs. GLDAS NOAH soil moisture 0-0.1m on number of days with comparison data.

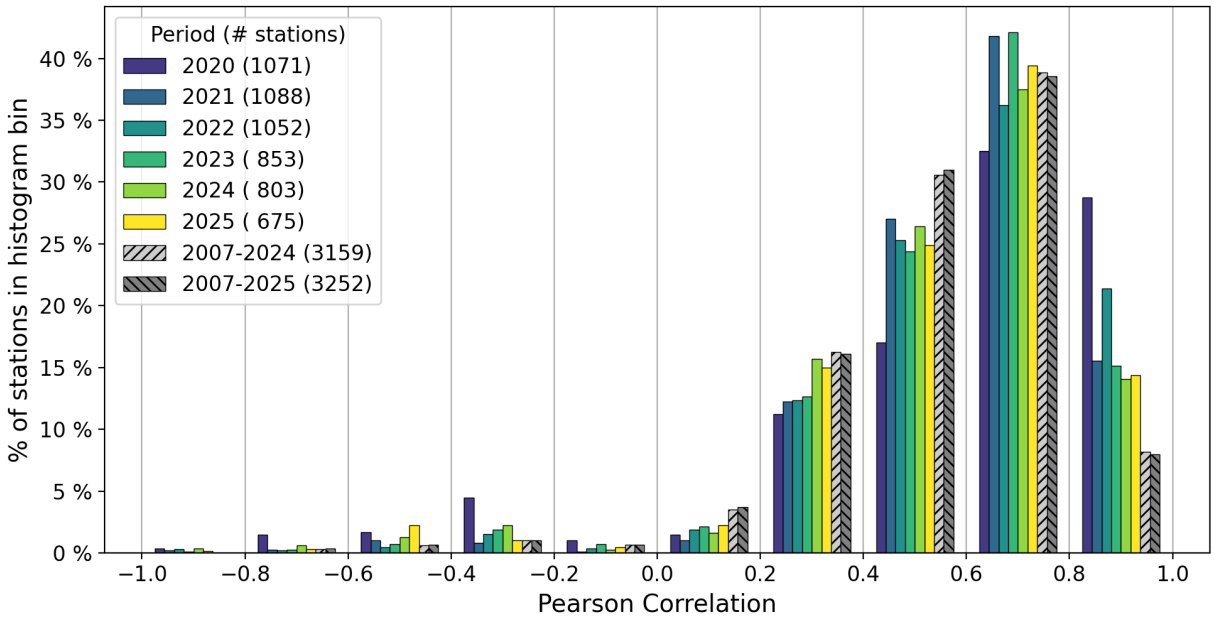


Figure 32: Comparison of single-year and multi-year Pearson Correlation of SWI T=1 vs. GLDAS NOAH soil moisture 0-0.1m. Each histogram bin represents the fraction of stations of the corresponding validation period that fall within the indicated interval (e.g. [-1, -0.8), … [0.8, 1)). The total number of available stations for each period is indicated in the legend.

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4.2.2.3 ubRMSD

An overall good performance is also observed with respect to ubRMSD in Figure 33 for the majority of in-situ stations with a median of ~0.061 m³/m³. 18.5% of in-situ stations meet the GCOS and CLMS goal ubRMSD <0.04m³/m³ and 89.5% meet the GCOS and CLMS threshold ubRMSD < 0.1m³/m³, see Figure 34 , which mirrors Figure 31 but for ubRMSD (and with similar conclusions to be drawn). Figure 35 presents a comparison of single-year ubRMSD values. More conspicuously than in the corresponding Pearson correlation diagram (Figure 32), the comparison with the multi-year diagrams suggests that, for the last 5 years, the agreement was better-than-average.

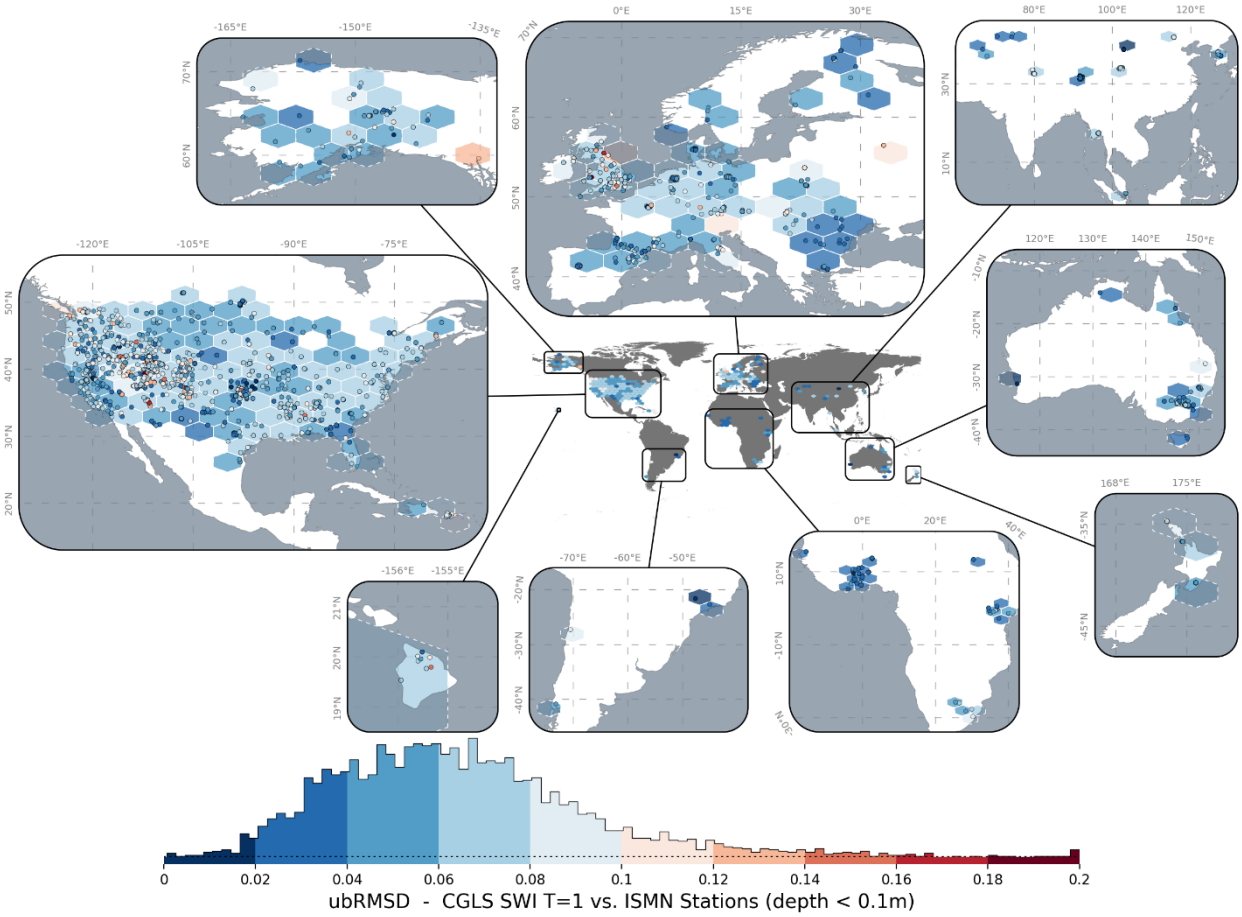


Figure 33: ubRMSD of CGLS SWI T=1 vs. ISMN station data (depth <0.1m)

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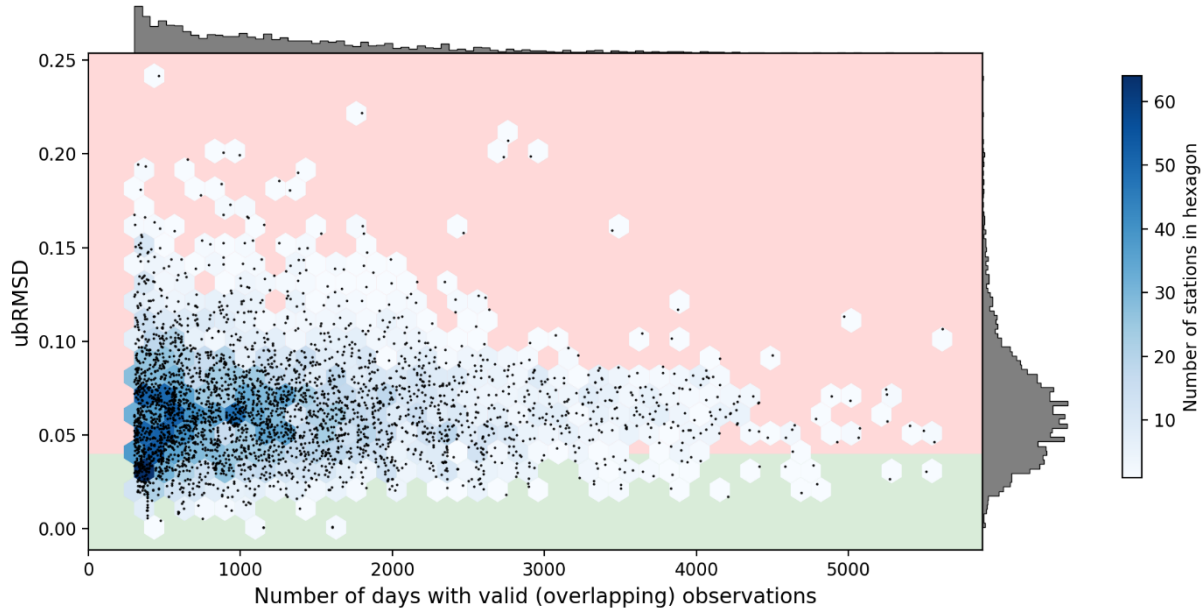


Figure 34: Dependency of ubRMSD of SWI T=1 vs. GLDAS NOAH soil moisture 0-0.1m on number of days with comparison data. Light-green shading indicates the region where the goal requirements are met.

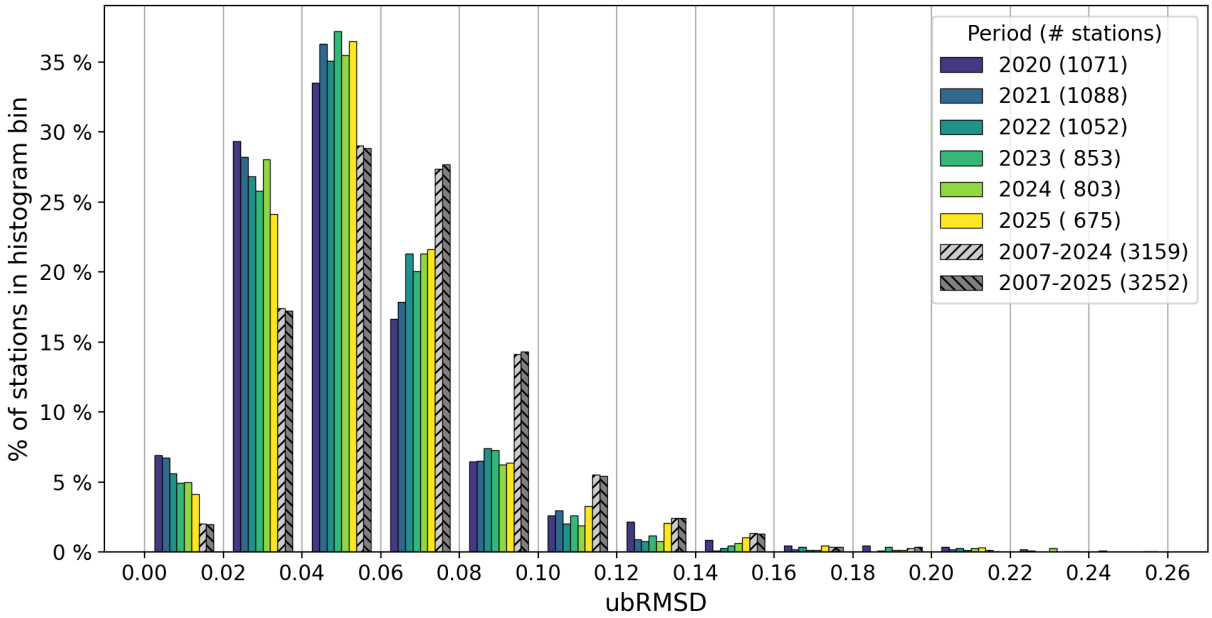


Figure 35: Comparison of single-year and multi-year ubRMSD of SWI T=1 vs. GLDAS NOAH soil moisture 0-0.1m. Each histogram bin represents the fraction of stations of the corresponding validation period that fall within the indicated interval (e.g. [0, 0.02), [0.02, 0.04), …). The total number of available stations for each period is indicated in the legend.

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5 CONCLUSIONS

5.1 COMPARISON BETWEEN SWIV3 AND SWIV4

The differences between SWIv4 and the predecessor SWIv3 were assessed by presenting an inter-comparison of the SWI values as well as differences in the obtained performance metrics against modelled and in-situ soil moisture for the overlap-period from 2007-01-01 to 2024-12-31.

Overall, a trend towards improved performance metrics was found for the SWIv4 dataset, highlighting the effectiveness of the implemented improvements in the new surface soil moisture input data stream (HSAF H29).

In the direct comparison, the most notable differences between the data versions are the changes in temporal masking and distinctly lower SWI averages for SWIv4 in the northern latitudes (above 45°) and large parts of East Asia.

5.2 ANNUAL VERIFICATION

The SWIv4 dataset was compared to modelled soil-moisture provided by GLDAS NOAH v2.1 as well as in-situ soil moisture observations provided by the ISMN network.

Table 7 shows in summary the compliance with the CLMS requirements for the SWI (cf. Section 2.1). Almost all (> 99%) of the grid points fulfil the threshold for the reference period when compared to GLDAS NOAH whereas the goal is fulfilled by 42-61%. Lower agreement is observed in the comparison to in-situ data: still above 90% with respect to the threshold requirement but down to 45-16% with respect to the goal. Interestingly, the latter category shows higher compliance figures for the current period than for the reference period whereas for almost all other figures in Table 7, it is the other way round.

No systematic pattern is visible with respect to a dependence of the statistics on the characteristic time length T.

Looking at the obtained correlation maps, one gains a differentiated picture of an overall good agreement of the SWIv4 dataset to both GLDAS NOAH and in-situ data, but (expected) weak performances in subarctic regions, densely vegetated areas (tropics), deserts, and areas that show high topographic complexity. All plots for the GLDAS comparison from Figure 22 to Figure 27 confirm the summary statistics in Table 7 that the current year showed less agreement than the reference period: the red colour (where “current” is inferior to “Reference”) prevails in all difference plots over most of the globe.

Table 7: Percentages of grid points for SWIv4.0.1 that meet the CLMS requirement goal/threshold for the reference period (2007-2024) and the current period (2025).

	CLMS goal (ubRMSD < 0.04m³/m³)		CLMS threshold (ubRMSD < 0.1 m³/m³)	
	GLDAS NOAH ref / cur	**ISMN** ref / cur	**GLDAS NOAH** ref / cur	**ISMN** ref / cur
SWI T=1	48 / 37	19 / 28	99 / 96	93 / 91
SWI T=20	42 / 31	16 / 30	99 / 92	91 / 94
SWI T=100	61 / 42	34 / 45	100 / 91	92 / 90

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6 RECOMMENDATIONS AND PERSPECTIVES

The annual verification of the SWIv4 product revealed a stable performance of the dataset with no detected inconsistencies. Therefore, a continued usage of the SWIv4 dataset is recommended.

As the comparison between the SWIv4 and its predecessor SWIv3 revealed improved performance for SWIv4, we recommend usage of the SWIv4 product. Still, we advise users who switch from SWIv3 to SWIv4 to assess potential impacts of the changes in temporal masking and the bias between SWIv3 and SWIv4 on the applications.

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